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A Mollified Itô–Tanaka Formula for Iterated Brownian Motion and Its First-Chaos Dual Closure

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Abstract

For iterated Brownian motion $Z_t = X_{|Y_t|}$ with two-sided Brownian scenery X and independent Brownian clock Y, we prove a mollified pathwise Itô–Tanaka formula

$$f(Z_t^\varepsilon) - f(Z_0^\varepsilon) \;=\; \delta_Y\big(B_{f,\varepsilon}(t)\big) \;+\; C_{Z,\varepsilon}(f;t),$$

holding at every $\varepsilon > 0$ on the product Wiener space, with expectation-level consistency $\lim_{\varepsilon\to 0}\mathbb{E}[C_{Z,\varepsilon}(f;t)] = u(t,x) - f(x)$ reproducing the fourth-order Allouba–Zheng Brownian-time PDE shadow. The unmollified pathwise correction does not survive in classical $L^2(\Omega)$: under a mild nondegeneracy hypothesis, the first-scenery-chaos projection has variance diverging as $\varepsilon \to 0$, and the obstruction is structural. Scalar Wick/Hermite subtraction removes the obvious diagonal contraction but does not target the identified first-chaos singularity; instead it introduces a different large first-chaos contribution and disrupts the Stein cancellation. The divergent first-chaos sector closes in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$, with an explicit limit kernel; this qualitative H^{-1} convergence is unconditional in $f \in \mathscr{S}(\mathbb{R})$. As supporting structure, a fixed-ε joint Wiener bidegree atlas locates the closure inside the chaos lattice and exhibits the pure-clock $\Pi_{0,0}$ block that carries the Brownian-time semigroup correction.

Keywords: iterated Brownian motion ; Itô formula ; Malliavin calculus ; obstruction ; first-chaos sector ; negative Sobolev closure ; Bouleau–Yor ; bidegree atlas.
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1 Introduction

Iterated Brownian motion

$$Z_t \;=\; X_{|Y_t|}, \qquad t \geq 0, \tag{1.1}$$

with two-sided Brownian scenery $X = (X_s)_{s\in\mathbb{R}}$ and an independent Brownian clock $Y = (Y_t)_{t\geq 0}$, is the canonical non-semimartingale composite-noise example in continuous

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stochastic calculus. In its natural filtration, Z is not Markov and not a semimartingale; it does not supply the finite predictable bracket structure required by the classical semimartingale Itô formula. Its averaged PDE shadow, identified by Allouba–Zheng [1], is the fourth-order Brownian-time equation

$$\partial_t u \;=\; \tfrac{1}{8}\,\partial_x^4 u \;+\; \frac{\partial_x^2 f}{2\sqrt{2\pi t}} \qquad \text{for } u(t,x)=\mathbb{E}[f(x+X_{|Y_t|})], \tag{1.2}$$

not the heat equation. The IBM observable correction operator is therefore $\frac{1}{8}\partial_x^4$ plus a boundary term, not a second-order infinitesimal generator.

Contributions and structure.

This paper proves three things.

(I) A mollified pathwise Itô–Tanaka formula (section 3). For $Z_t^\varepsilon = x + X^\varepsilon_{|Y_t|}$ with $X^\varepsilon = \rho_\varepsilon * X$, $f \in \mathscr{S}(\mathbb{R})$, and $g_\varepsilon(r) = f(x + X_r^\varepsilon)$,

$$f(Z_t^\varepsilon) - f(Z_0^\varepsilon) \;=\; \delta_Y\big(B_{f,\varepsilon}(t)\big) \;+\; C_{Z,\varepsilon}(f;t), \tag{1.3}$$

holds pathwise for every $\varepsilon > 0$ (theorem 3.5), with clock-Skorohod integrand $B_{f,\varepsilon}(t)(s) = g_\varepsilon'(|Y_s|)\,\mathrm{sgn}(Y_s)\mathbf{1}_{[0,t]}(s)$ and mollified pathwise correction $C_{Z,\varepsilon}(f;t) = g_\varepsilon'(0)L_t^0(Y) + \frac{1}{2}\int_0^t g_\varepsilon''(|Y_s|)\,ds$. In expectation, the correction reproduces the Allouba–Zheng PDE shadow: $\lim_{\varepsilon\to 0}\mathbb{E}[C_{Z,\varepsilon}(f;t)] = u(t,x) - f(x)$ (theorem 3.6). The proofs use joint Malliavin calculus on the product Wiener space $\Omega_X \times \Omega_Y$.

(II) The $L^2(\Omega)$ obstruction (section 4). Under the nondegeneracy hypothesis $(P_\eta f')(x) \not\equiv 0$ on $(0,\infty)$, the centred first-scenery-chaos projection of $C_{Z,\varepsilon}(f;t)$ has variance

$$\liminf_{\varepsilon\to 0}\ \mathrm{Var}\big(\Pi_1^X\, C_{Z,\varepsilon}(f;t)\big) \;=\; +\infty \tag{1.4}$$

(theorem 4.4). The argument exhibits a Stein cancellation of leading ε^{-1} terms across the two summands of g_ε'', leaving a residual ρ_ε'-kernel whose L^2-divergence is forced by the Bouleau–Yor spatial quadratic variation of the Brownian clock local time. The obstruction is decisive at the $L^2(\Omega)$ level and is structurally Wick/Hermite-resistant: scalar Wick-square subtraction removes the diagonal contraction in $\mathcal{H}_0^X \oplus \mathcal{H}_2^X$ but does not target the T_3 first-chaos singularity carried by $W_r = \partial_r^2 X^\varepsilon \in \mathcal{H}_1^X$, and it introduces a large $\xi_{r,\varepsilon}$ counterterm that disrupts the Stein cancellation (section A).

(III) First-chaos dual closure (section 5). The divergent first-chaos sector closes in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$: the conditional Malliavin kernel K_ε admits an interior/boundary decomposition and converges qualitatively to $K = L_t^0(Y)\,f'(x)\,\delta_0 + K^{\mathrm{int}}$ (theorem 5.13); this qualitative H^{-1} convergence is unconditional in $f \in \mathscr{S}(\mathbb{R})$, while a quantitative $\varepsilon^{1/2-\eta}$ rate is conditional on a Marcus–Rosen-type moment assumption (theorems 5.8 and 5.11). The limit kernel is generically only H^{-1}, so the limit object is a generalized Y-conditioned first-chaos functional defined by duality against H^1 test kernels (theorem 5.1), not an ordinary $L^2(\Omega)$ random variable.

We additionally provide, for context, a fixed-ε structural map of $C_{Z,\varepsilon}(f;t)$ across the joint Wiener bidegree lattice $\mathcal{H}_m^X \otimes \mathcal{H}_n^Y$ (section 6). At fixed ε the pure-clock $\Pi_{0,0}$ sector recovers the Brownian-time semigroup correction (theorem 6.3, by reference to theorem 3.6); the $(1,n)$ slices route to the dual closure block-wise (theorem 6.5); and the $(2,0)$ sector admits a six-channel energy-contraction decomposition (theorem 6.8). This atlas is supporting structure for the main result (III), not a fourth pillar: cross-sector $\varepsilon \to 0$ summability is open (theorem 7.1).

The operator-derivative bridge.

The dual object for the singular first-chaos sector is named via a minimal operator-derivative bridge, introduced as the opening subsection of section 5 (see section 5). The abstract framework (operator derivative as the Hilbert adjoint of the divergence on an energy Hilbert space, with the chain rule following from a cylindrical reduction) is developed in [4] and used only contextually here. The first-chaos dual closure of section 5 and the fixed-ε atlas of section 6 are stated and proved using joint Malliavin calculus on the product Wiener space; the bridge identifies the negative-Sobolev block $H^{-1}(\mathbb{R})$ as the test/dual space, but the proofs of theorems 5.13, 6.3, 6.5 and 6.8 are self-contained.

The paper is organised as follows. Section 2 fixes the joint Wiener setup and the mollifier estimates. Section 3 proves the mollified Itô–Tanaka identity and its fourth-order expectation-level consistency (the only statement of the expectation–semigroup link). Section 4 proves the first-chaos $L^2(\Omega)$ obstruction. Section 5 proves the first-chaos dual closure in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$; an opening subsection (section 5) names the dual object via a minimal operator-derivative bridge and records the density reframe (F3). Section 6 maps the bidegree atlas at fixed $\varepsilon > 0$. Section 7 collects the three open problems and directions for future work; two appendices record the Wick/Hermite analysis and the Fourier-mode PDE-shadow diagnostic.

2 Setup

This short section fixes the joint Wiener setup, the mollifier, and the quantitative estimates that the rest of the paper draws on.

Source data, mollifier, and joint chaos.

Let $X = (X_z)_{z\in\mathbb{R}}$ be a two-sided Brownian scenery with variance $\mathrm{Var}(X_z) = |z|$ and covariance $\mathrm{Cov}(X_z, X_{z'}) = \frac{1}{2}(|z| + |z'| - |z - z'|)$, and let $Y = (Y_t)_{t\geq 0}$ be an independent one-sided Brownian clock with $Y_0 = 0$. We write $\mathcal{F}_X = \sigma(X)$ and $\mathcal{F}_Y = \sigma(Y)$ for the marginal σ-algebras, Ω_X and Ω_Y for the corresponding marginal probability spaces, and $\mathbb{E}_X$, $\mathbb{E}_Y$ for the partial expectations integrating out the respective Gaussian factor. We fix a test function $f \in \mathscr{S}(\mathbb{R})$ and a basepoint $x \in \mathbb{R}$ throughout.

Fix a mollifier $\rho \in C_c^\infty(\mathbb{R})$ that is even and non-negative with $\int \rho = 1$ and $\operatorname{supp}\rho \subseteq [-c, c]$, and set $\rho_\varepsilon(\cdot) = \varepsilon^{-1}\rho(\cdot/\varepsilon)$. The *mollified scenery* is $X^\varepsilon := \rho_\varepsilon * X$ and the *mollified IBM* is

$$Z_t^\varepsilon \;:=\; x + X^\varepsilon_{|Y_t|}. \tag{2.1}$$

We write $U_r := X_r^\varepsilon$ for the value of the mollified scenery at the time-shift r, and $V_r := \partial_r U_r = (\rho_\varepsilon' * X)(r)$ and $W_r := \partial_r^2 U_r = (\rho_\varepsilon'' * X)(r)$ for its first and second spatial derivatives. The pointwise variance of U_r is denoted $\sigma_\varepsilon^2(r) := \mathrm{Var}(U_r)$, and the variance of V_r is $\sigma_V^2 := \mathrm{Var}(V_r)$, independent of r in the bulk by theorem 2.3. For the test function f at basepoint x we set $g_\varepsilon(r) := f(x + U_r)$, with derivatives in r given by the chain rule:

$$g_\varepsilon'(r) = f'(x + U_r)V_r, \qquad g_\varepsilon''(r) = f''(x + U_r)V_r^2 + f'(x + U_r)W_r. \tag{2.2}$$

We write $L_t^x(Y)$ for the Brownian local time of Y at level x up to time t, and $L_t^x(|Y|)$ for the spatial local time of the reflected motion $|Y|$.

The product Wiener space $\Omega_X \times \Omega_Y$ admits the joint Wiener chaos decomposition

$$L^2(\Omega_X \times \Omega_Y) \;=\; \bigoplus_{m,n\geq 0} \mathcal{H}_m^X \otimes \mathcal{H}_n^Y, \tag{2.3}$$

where $\mathcal{H}_m^X \subset L^2(\Omega_X)$ and $\mathcal{H}_n^Y \subset L^2(\Omega_Y)$ are the m-th scenery and n-th clock Wiener chaoses. We write Π_m^X, Π_n^Y for the orthogonal projections onto $\mathcal{H}_m^X$ and $\mathcal{H}_n^Y$, extended

to the joint space as $\Pi_m^X \otimes \mathrm{Id}_Y$ and $\mathrm{Id}_X \otimes \Pi_n^Y$ respectively. These commute, and we write $\Pi_{m,n} = \Pi_m^X \Pi_n^Y$ for the bidegree-(m,n) projection. Two identities are used below: $\Pi_0^Y = \mathbb{E}_Y$ is the conditional expectation that integrates out the clock, and $\Pi_0^X = \mathbb{E}_X$ integrates out the scenery.

The partial Malliavin derivatives D_X, D_Y act on each Gaussian factor; their adjoints are the Skorohod divergences $\delta_X = D_X^*$ and $\delta_Y = D_Y^*$ [7, 5]. The scenery X admits the white-noise representation $X_z = \int_{\mathbb{R}} \sigma_z(w)\, d\xi(w)$ with the kernel $\sigma_z(w) = \mathrm{sgn}(z)\mathbf{1}_{[0,z]_*}(w)$. Differentiating under the convolution gives the Malliavin derivative kernels of the mollified scenery,

$$D_w U_r = \xi_{r,\varepsilon}(w), \quad D_w V_r = \rho_\varepsilon(r-w), \quad D_w W_r = \rho_\varepsilon'(r-w), \tag{2.4}$$

where $\xi_{r,\varepsilon}(w) \coloneqq \int \rho_\varepsilon(r-z)\sigma_z(w)\, dz$.

Quantitative estimates.

We collect the explicit second-moment bounds used throughout, and record the bulk covariance identities that drive the Stein cancellation underlying both the obstruction (section 4) and the dual closure (section 5). All constants depend on ρ only via c, $\|\rho\|_\infty$, $\int |v|\rho(v)\, dv$, $\int \rho^2$, $\int (\rho')^2$.

Lemma 2.1 (Scenery mollifier L^2-rate, uniform in x)**.** *For every $x \in \mathbb{R}$,*

$$\mathbb{E}\big[(X_x^\varepsilon - X_x)^2\big] \;\le\; C_\rho\, \varepsilon, \qquad C_\rho \;\coloneqq\; \int_{\mathbb{R}} |v|\rho(v)\, dv, \tag{2.5}$$

hence $\mathbb{E}|X_x^\varepsilon - X_x| \le C_\rho^{1/2}\varepsilon^{1/2}$ uniformly in x. In particular $X_0^\varepsilon \to 0$ in L^2 as $\varepsilon \to 0$ since $X_0 = 0$ a.s.

Proof. For two-sided Brownian scenery, $\mathbb{E}[(X_z - X_x)^2] = |z-x|$ for every $z, x \in \mathbb{R}$: when z, x have the same sign or one is zero, this is the standard one-sided Brownian increment variance; when $z > 0 > x$, independence of X_z and X_x gives $\mathbb{E}[(X_z - X_x)^2] = \mathbb{E}X_z^2 + \mathbb{E}X_x^2 = z + |x| = z - x = |z-x|$ (and symmetrically for $x > 0 > z$). Writing $X_x^\varepsilon - X_x = \int_{\mathbb{R}} \rho_\varepsilon(x-z)(X_z - X_x)\, dz$ and applying Jensen with the probability measure $\rho_\varepsilon(x-z)\, dz$, $\mathbb{E}[(X_x^\varepsilon - X_x)^2] \le \int_{\mathbb{R}} \rho_\varepsilon(x-z)|z-x|\, dz = \int_{\mathbb{R}} \rho(v)|v|\, \varepsilon\, dv = C_\rho \varepsilon$, uniform in x. □

Lemma 2.2 (Scenery-derivative kernel: L^2 bound and rate)**.** *With $H_\varepsilon(y,\xi) \coloneqq (D_X X_y^\varepsilon)(\xi) = (\rho_\varepsilon * \sigma_\cdot(\xi))(y)$:*

- *(a) $\|H_\varepsilon(y,\cdot)\|_{L^2(\mathbb{R})}^2 \le |y| + C_\rho\varepsilon$ for every $y \in \mathbb{R}$.*
- *(b) $\|H_\varepsilon(y,\cdot) - \sigma_y\|_{L^2(\mathbb{R})}^2 \le C_\rho'\varepsilon$ for every $y \in \mathbb{R}$, with C_ρ' depending only on $\operatorname{supp}\rho$.*

Proof. (a) Jensen and $\|\sigma_z\|_{L^2(\mathbb{R})}^2 = |z|$ give $\int_{\mathbb{R}} H_\varepsilon(y,\xi)^2\, d\xi \le \int_{\mathbb{R}} \rho_\varepsilon(y-z)|z|\, dz \le |y| + C_\rho\varepsilon$. (b) The difference vanishes off the $O(\varepsilon)$-neighbourhood of the jump locus of σ_y and is uniformly bounded by 1 there; the $L^2(\mathbb{R}, d\xi)$ mass is $O(\varepsilon)$ uniformly in y. □

Lemma 2.3 (Spatial derivative of the mollified scenery)**.** *For $|y| > c\varepsilon$, $V_y = (\rho_\varepsilon' * X)(y) = \int_{\mathbb{R}} \rho_\varepsilon(y-z)\, dX_z$ is centred Gaussian with*

$$\mathrm{Var}(V_y) \;=\; \frac{c_\rho^{(1)}}{\varepsilon}, \qquad c_\rho^{(1)} \coloneqq \int_{\mathbb{R}} \rho(v)^2\, dv. \tag{2.6}$$

Jensen and (2.6) extend to give $\mathbb{E}[V_y^2] \le c_\rho^{(1)}/\varepsilon$ uniformly in $y \in \mathbb{R}$. Similarly, in the bulk $\mathrm{Var}(W_y) = \int_{\mathbb{R}} \rho_\varepsilon'(y-z)^2\, dz = c_\rho^{(2)}\varepsilon^{-3}$ with $c_\rho^{(2)} = \int (\rho')^2$.

Proof. For $|y| > c\varepsilon$, $z \mapsto \rho_\varepsilon(y - z)$ is supported entirely on one side of 0, so $\partial_y X^\varepsilon(y) = \int \rho_\varepsilon(y - z)\, dX_z$ is an Itô integral of an L^2 integrand; the Itô isometry yields $\mathrm{Var}(V_y) = \|\rho_\varepsilon\|_{L^2}^2 = \int \rho^2/\varepsilon$. The boundary layer $|y| \le c\varepsilon$ is handled by Jensen on the same kernel. □

Finally, for $|r| > c\varepsilon$ the bulk covariance identities

$$(\sigma_\varepsilon^2)'(r) = 1, \qquad \mathrm{Cov}(U_r, V_r) = \tfrac{1}{2}, \qquad \sigma_V^2 + \mathrm{Cov}(U_r, W_r) = 0 \tag{2.7}$$

hold, the last by differentiating $2\,\mathrm{Cov}(U_r, V_r) = (\sigma_\varepsilon^2)'(r) = 1$ once more; we use these without re-proof. The boundary-layer corrections in $r \in [0, c\varepsilon]$ are quantified in section 5 (theorems 5.5 and 5.7).

3 The mollified pathwise Itô–Tanaka formula

With the joint Wiener setup of section 2 in hand, we now prove the pathwise Itô–Tanaka identity for the mollified IBM and identify its expectation-level fourth-order consistency. We begin with the deterministic Brownian-time semigroup correction, which the mollified correction must reproduce in expectation as $\varepsilon \to 0$; the mollified pathwise identity follows by applying the classical Itô–Tanaka formula to $g_\varepsilon(|Y_t|)$ in the Brownian-time variable; and the expectation-level consistency assembles the two. The expectation-level statement (theorem 3.6) is the only locus in the paper that links the pathwise correction to the Brownian-time semigroup; the atlas of section 6 refers back to it.

The Brownian-time semigroup correction.

Before the pathwise formula, we record the expectation-level correction identified by Allouba–Zheng. The mollified pathwise identity will reproduce this correction in the $\varepsilon \to 0$ limit (theorem 3.6 below).

Theorem 3.1 (Brownian-time semigroup correction)**.** *For $f \in \mathscr{S}(\mathbb{R})$, set $u(t,x) := \mathbb{E}[f(x + X_{|Y_t|})]$. Then $u \in C^\infty((0,\infty)\times\mathbb{R})$, $u(0+,x) = f(x)$, and for every $t > 0$,*

$$\partial_t u(t,x) \;=\; \frac{1}{8}\,\partial_x^4 u(t,x) \;+\; \frac{1}{2\sqrt{2\pi t}}\,\partial_x^2 f(x). \tag{3.1}$$

Proof. Conditioning on $|Y_t|$ and using independence of X and Y, $u(t,x) = \int_0^\infty (P_s f)(x)\, p_t(s)\, ds$ with P_s the heat semigroup of X (variance s, so $\partial_s P_s f = \frac{1}{2}\partial_x^2 P_s f$) and $p_t(s) = \sqrt{2/(\pi t)}\, e^{-s^2/(2t)}$ the half-normal density, satisfying $\partial_t p_t = \frac{1}{2}\partial_s^2 p_t$ pointwise for $s > 0$, $\partial_s p_t(0) = 0$, $p_t(0) = \sqrt{2/(\pi t)}$. Hence

$$\partial_t u(t,x) \;=\; \tfrac{1}{2}\int_0^\infty (P_s f)(x)\, \partial_s^2 p_t(s)\, ds.$$

Integrate by parts twice in s; boundary terms at $s = \infty$ vanish:

$$\begin{aligned}
\tfrac{1}{2}\int_0^\infty (P_s f)(x)\, \partial_s^2 p_t(s)\, ds &= -\tfrac{1}{2}\int_0^\infty \partial_s (P_s f)(x)\, \partial_s p_t(s)\, ds \quad (\partial_s p_t(0) = 0) \\
&= -\tfrac{1}{4}\int_0^\infty \partial_x^2 P_s f(x)\, \partial_s p_t(s)\, ds \\
&= -\tfrac{1}{4}\big[\partial_x^2 P_s f(x)\, p_t(s)\big]_0^\infty + \tfrac{1}{4}\int_0^\infty \partial_s \partial_x^2 P_s f(x)\, p_t(s)\, ds \\
&= \tfrac{1}{4}\,\partial_x^2 f(x)\, p_t(0) + \tfrac{1}{8}\int_0^\infty \partial_x^4 P_s f(x)\, p_t(s)\, ds.
\end{aligned}$$

The boundary term equals $\partial_x^2 f(x)/(2\sqrt{2\pi t})$; the remaining integral equals $\frac{1}{8}\partial_x^4 u(t,x)$. Smoothness on $(0,\infty)\times\mathbb{R}$ and the initial condition follow from standard regularising properties of G_s and p_t. □

Definition 3.2 (Semigroup correction density and integrated correction)**.** *For $f \in \mathscr{S}(\mathbb{R})$ and $t > 0$,*

$$\mathcal{C}_Z^{\mathrm{sg}}(f;t,x) \;\coloneqq\; \frac{1}{8}\,\partial_x^4 u(t,x) \;+\; \frac{1}{2\sqrt{2\pi t}}\,\partial_x^2 f(x), \quad C_Z^{\mathrm{sg}}(f;t,x) \;\coloneqq\; \int_0^t \mathcal{C}_Z^{\mathrm{sg}}(f;s,x)\,ds. \tag{3.2}$$

The integrand is integrable on $(0,t]$: $\int_0^t s^{-1/2}\,ds = 2\sqrt{t} < \infty$.

Corollary 3.3 (Integrated semigroup form)**.** *For $f \in \mathscr{S}(\mathbb{R})$ and $t > 0$, $u(t,x) - f(x) = C_Z^{\mathrm{sg}}(f;t,x)$.*

Proof. Integrate (3.1) from 0 to t and use $u(0+,x) = f(x)$. □

The mollified pathwise Itô–Tanaka identity.

We turn now to the pathwise formula at every $\varepsilon > 0$. Conditionally on the scenery X, the function $r \mapsto g_\varepsilon(r) = f(x + X_r^\varepsilon)$ is smooth, so the classical Itô–Tanaka formula applied to $g_\varepsilon(|Y_t|)$ in the Brownian-time variable yields the identity below.

Proposition 3.4 (Mollified pathwise Itô–Tanaka identity)**.** *For every $\varepsilon > 0$, $t \geq 0$, $f \in \mathscr{S}(\mathbb{R})$,*

$$f(Z_t^\varepsilon) - f(Z_0^\varepsilon) \;=\; M_{\varepsilon,t}^Y \;+\; g_\varepsilon'(0)\,L_t^0(Y) \;+\; \tfrac{1}{2}\int_0^t g_\varepsilon''(|Y_s|)\,ds, \tag{3.3}$$

where $M_{\varepsilon,t}^Y = \int_0^t g_\varepsilon'(|Y_s|)\,\mathrm{sgn}(Y_s)\,dY_s$ is a continuous L^2-martingale in Y (conditional on X).

Proof. Tanaka's formula gives $d|Y_t| = \mathrm{sgn}(Y_t)\,dY_t + dL_t^0(Y)$ and $\langle |Y| \rangle_t = t$. For each ω_X, $g_\varepsilon \in C^\infty(\mathbb{R})$, so the Itô–Tanaka formula applied to $g_\varepsilon(|Y_t|)$ yields, using $g_\varepsilon'(|Y_s|)\,dL_s^0 = g_\varepsilon'(0)\,dL_s^0$ (since dL_s^0 is supported on $\{Y_s = 0\}$),

$$g_\varepsilon(|Y_t|) - g_\varepsilon(0) = \int_0^t g_\varepsilon'(|Y_s|)\,\mathrm{sgn}(Y_s)\,dY_s + g_\varepsilon'(0) L_t^0(Y) + \tfrac{1}{2}\int_0^t g_\varepsilon''(|Y_s|)\,ds.$$

The left-hand side equals $f(Z_t^\varepsilon) - f(Z_0^\varepsilon)$ since $Z_t^\varepsilon = x + X_{|Y_t|}^\varepsilon$ and $|Y_0| = 0$. The explicit formulas for g_ε', g_ε'' in (2.2) are the chain rule. □

Define the *mollified pathwise correction*

$$C_{Z,\varepsilon}(f;t) \;\coloneqq\; g_\varepsilon'(0)\,L_t^0(Y) \;+\; \tfrac{1}{2}\int_0^t g_\varepsilon''(|Y_s|)\,ds. \tag{3.4}$$

Theorem 3.5 (Mollified IBM Itô formula)**.** *For every $\varepsilon > 0$, $t \geq 0$, and $f \in \mathscr{S}(\mathbb{R})$:*

1. *(Pathwise identity)*

$$f(Z_t^\varepsilon) - f(Z_0^\varepsilon) \;=\; \delta_Y\big(B_{f,\varepsilon}(t)\big) \;+\; C_{Z,\varepsilon}(f;t),$$

   with $B_{f,\varepsilon}(t)(s) = g_\varepsilon'(|Y_s|)\,\mathrm{sgn}(Y_s)\mathbf{1}_{[0,t]}(s)$ and $C_{Z,\varepsilon}$ as in (3.4).
2. *(Skorohod–Itô identification) The integrand $B_{f,\varepsilon}(t) \in L^2(\Omega; L^2(\mathbb{R}_+))$, the clock Skorohod divergence on the Y-adapted integrand coincides with the Itô integral $M_{\varepsilon,t}^Y$, and $\mathbb{E}[\delta_Y(B_{f,\varepsilon}(t))] = 0$.*

Proof. Part (1) is theorem 3.4 repackaged. For (2), $B_{f,\varepsilon}(t)$ is Y-adapted with continuous paths in s. By theorem 2.3, $\mathbb{E}[V_r^2] \le c_\rho^{(1)}/\varepsilon$ uniformly in r, so

$$\mathbb{E}\int_0^t B_{f,\varepsilon}(t)(s)^2\,ds \;\le\; \|f'\|_\infty^2\, t\, c_\rho^{(1)}/\varepsilon \;<\; \infty.$$

The source clock Skorohod divergence on Y-adapted L^2 integrands coincides with the Itô integral [7, Prop. 1.3.4]. □

Taking expectation in (3.3) eliminates the martingale term and the boundary local-time piece (the latter by a reflection-symmetry argument, theorem 3.7), leaving the expectation-level identification of $\mathbb{E}[C_{Z,\varepsilon}]$ as the Brownian-time semigroup increment. This is the only locus in the paper that links the pathwise correction to the Allouba–Zheng semigroup; the bidegree atlas of section 6 will refer back to it, not re-prove it.

Theorem 3.6 (Expectation-level consistency)**.** *For every $f \in \mathscr{S}(\mathbb{R})$, $\varepsilon > 0$, and $t > 0$,*

$$\mathbb{E}[C_{Z,\varepsilon}(f;t)] \;=\; U^\varepsilon(t,x) - v^\varepsilon(0,x), \tag{3.5}$$

where $v^\varepsilon(r,x) = \mathbb{E}_X[f(x+U_r)]$ and $U^\varepsilon(t,x) = \mathbb{E}[f(x+U_{|Y_t|})] = \int_0^\infty v^\varepsilon(r,x)\,p_t(r)\,dr$. Consequently

$$\lim_{\varepsilon\to 0}\mathbb{E}[C_{Z,\varepsilon}(f;t)] \;=\; u(t,x) - f(x) \;=\; C_Z^{\mathrm{sg}}(f;t,x), \tag{3.6}$$

with $u(t,x) = \mathbb{E}[f(x+X_{|Y_t|})]$ and C_Z^{sg} from theorem 3.2.

Proof. Take expectation in (3.3): $\mathbb{E}[M^Y_{\varepsilon,t}] = 0$, so $\mathbb{E}[f(Z^\varepsilon_t) - f(Z^\varepsilon_0)] = \mathbb{E}[C_{Z,\varepsilon}(f;t)]$. By definition $\mathbb{E}[f(Z^\varepsilon_t)] = U^\varepsilon(t,x)$ and $\mathbb{E}[f(Z^\varepsilon_0)] = v^\varepsilon(0,x)$, giving (3.5). For (3.6): Lipschitz f and theorem 2.1 give $\mathbb{E}|f(x+X^\varepsilon_r) - f(x+X_r)| \le \|f'\|_\infty C_\rho^{1/2}\varepsilon^{1/2}$ uniformly in r; averaging against the law of $|Y_t|$ yields $U^\varepsilon(t,x) \to u(t,x)$. At $r=0$, $X^\varepsilon_0 \to 0$ in L^2 (theorem 2.1 with $X_0 = 0$), so $v^\varepsilon(0,x) \to f(x)$ by Lipschitz f. The second equality is theorem 3.3. □

Remark 3.7 (Local-time term is expectation-invisible)**.** The boundary local-time term is expectation-invisible: a reflection-symmetry argument gives $\mathbb{E}[g'_\varepsilon(0)L^0_t(Y)] = 0$ for every $\varepsilon > 0$ and $t > 0$. Indeed, $L^0_t(Y)$ is Y-measurable and $g'_\varepsilon(0)$ is X-measurable, so by independence $\mathbb{E}[g'_\varepsilon(0)L^0_t(Y)] = \mathbb{E}_X[g'_\varepsilon(0)]\,\mathbb{E}_Y[L^0_t(Y)]$; under the reflection $X_z \leftrightarrow X_{-z}$, X^ε_0 is invariant (even integrand against even ρ_ε) while $V_0 = -\int \rho'_\varepsilon(z)X_z\,dz$ picks up a sign (odd integrand), so $\mathbb{E}_X[h(X^\varepsilon_0)V_0] = 0$ for every Borel h, and in particular $\mathbb{E}_X[g'_\varepsilon(0)] = 0$. The full expectation in (3.5) is therefore carried by the interior g''_ε-integral; the same vanishing identity reappears in theorem 6.3 via the boundary covariance theorem 5.5.

The correction is fourth-order in expectation rather than in the clock variable. The clock Itô formula behind (3.3) is second order in the Brownian-time variable: it is the ordinary Itô–Tanaka formula applied to $g_\varepsilon(|Y_t|)$. The coefficients g'_ε, g''_ε are however first and second spatial derivatives of a Brownian scenery, so the PDE shadow of $\mathbb{E}[C_{Z,\varepsilon}]$ is fourth order in the spatial variable: $\frac{1}{8}\partial_x^4 u$ is the shadow of the $f'(x+U_r)W_r$ contribution to g''_ε after averaging over the half-normal clock distribution, and $\partial_x^2 f/(2\sqrt{2\pi s})$ is its half-normal boundary correction. Classical $\langle\cdot\rangle$-tracking sees only the second-order clock term and misses the fourth-order spatial shadow that is the actual Itô correction for IBM.

4 The $L^2(\Omega)$ obstruction

The mollified formula (1.3) is an identity at every $\varepsilon > 0$, and its expectation reproduces the Allouba–Zheng PDE shadow as $\varepsilon \to 0$. The next question is whether the pathwise correction itself converges as $\varepsilon \to 0$ in $L^2(\Omega)$. It does not: the centred fluctuation $\tilde{C}_{Z,\varepsilon} := C_{Z,\varepsilon} - \mathbb{E}[C_{Z,\varepsilon}]$ has variance diverging as $\varepsilon \to 0$ (theorem 4.4 below). The

argument exhibits a Stein cancellation of leading ε^{-1} terms in the Malliavin kernel and isolates a residual ρ'_ε-kernel whose L^2-divergence is forced by the Bouleau–Yor spatial quadratic variation of the Brownian clock local time. The obstruction is decisive at the $L^2(\Omega)$ level and is Wick/Hermite-resistant (section A).

We first record three auxiliary lemmas: the Brownian local-time Sobolev failure that drives the divergence, a conditional Stroock projection identity that exhibits the first-chaos Malliavin kernel, and a Plancherel-type blow-up criterion for L^2-approximants of non-H^1 limits. The obstruction theorem then follows.

Lemma 4.1 (Brownian local-time Sobolev regularity)**.** *Let Y be standard Brownian motion, $t > 0$, and write $L := L_t(|Y|)$ for the spatial local time of $|Y|$ on $[0,t]$, supported in $[0, M_t]$ with $M_t = \sup_{[0,t]} |Y_s| > 0$ a.s.*

1. *For every $s < \frac{1}{2}$, $L \in H^s_{\mathrm{loc}}(\mathbb{R})$ a.s.*
2. *Fix any compact $J = [a,b] \subset (0,\infty)$. On the event $\{M_t > b\}$ (positive $\mathbb{P}_Y$-probability), the spatial occupation of J is strictly positive, hence the Bouleau–Yor spatial quadratic variation $\mathrm{QV}_J(L) > 0$ a.s., and $L \notin H^1_{\mathrm{loc}}(J)$ a.s. on $\{M_t > b\}$.*

Proof. By Tanaka, $|Y_t| = \int_0^t \mathrm{sgn}(Y_s)\, dY_s + L^0_t(Y)$ is a continuous semimartingale. (i) $r \mapsto L^r_t(|Y|)$ is $1/2^-$ Hölder continuous ([8], Ch. VI, Theorem VI.1.7 for Y, applied via Tanaka to $|Y|$), so $L \in H^s_{\mathrm{loc}}(\mathbb{R})$ for every $s < 1/2$. (ii) The occupation identity $\int_J L^r_t(|Y|)\, dr = \int_0^t \mathbf{1}_J(|Y_s|)\, ds > 0$ on $\{M_t > b\}$ is the standard occupation-time formula ([8], Ch. VI). Bouleau–Yor [3] applied to the continuous semimartingale $|Y|$ yields $\mathrm{QV}_J(L) > 0$ a.s. on the same event. If $L \in H^1_{\mathrm{loc}}(J)$, L is absolutely continuous on J with $L' \in L^2(J)$, and Cauchy–Schwarz gives the spatial quadratic variation along refining partitions bounded by $|\Delta|\, \|L'\|^2_{L^2(J)} \to 0$; combined with $\mathrm{QV}_J(L) > 0$, this is a contradiction. □

Lemma 4.2 (Conditional first-scenery-chaos projection)**.** *For $F \in \mathbb{D}^{1,2}(\Omega_X \times \Omega_Y)$ in the scenery direction,*

$$\Pi^X_1 F \;=\; \delta_X\big(K_F(\cdot\,;Y)\big), \qquad K_F(w;Y) \;:=\; \mathbb{E}_X[D^X_w F \mid Y], \tag{4.1}$$

with $K_F(\cdot;Y) \in L^2(\mathbb{R}, dw)$ a.s.

Proof. For smooth cylindrical F, the section $F(\cdot,\omega_Y)$ is smooth cylindrical on the scenery factor; Stroock on X ([7, Prop. 1.3.1]) gives the fibrewise identity. Taking $\mathbb{E}_X[\,\cdot \mid Y]$ and applying δ_X produces (4.1) on the cylindrical subspace. For general $F \in \mathbb{D}^{1,2}$ in the scenery direction, approximate by smooth cylindrical F_n in $\mathbb{D}^{1,2}$ ([7, §1.2]); L^2-contraction of conditional expectation gives $K_{F_n} \to K_F$ in $L^2(\Omega; L^2(\mathbb{R}))$; chaos orthogonality gives $\Pi^X_1 F_n \to \Pi^X_1 F$ in L^2; Itô isometry gives $\delta_X K_{F_n} \to \delta_X K_F$ in L^2; (4.1) passes to the limit. □

Theorem 4.2 applies to $C_{Z,\varepsilon}(f;t)$ for each $\varepsilon > 0$: g'_ε, g''_ε are smooth functionals of the finite Gaussian collection (U_r, V_r, W_r), and the Schwartz bounds on f together with Fubini over $s \in [0,t]$ place $C_{Z,\varepsilon} \in \mathbb{D}^{1,2}$ in the scenery direction.

Lemma 4.3 (Sobolev gradient blow-up under L^2 approximation)**.** *Let $u_\varepsilon \to u$ in $L^2(\mathbb{R})$, with each $u_\varepsilon \in C^1(\mathbb{R}) \cap H^1(\mathbb{R})$, and $u \notin H^1(\mathbb{R})$. Then $\liminf_{\varepsilon\to 0} \|u'_\varepsilon\|_{L^2(\mathbb{R})} = +\infty$.*

Proof. By Plancherel, $\|u'_\varepsilon\|^2_{L^2} = (2\pi)^{-1} \int k^2 |\widehat{u_\varepsilon}|^2\, dk$. For any $R > 0$, dominated convergence on $|k| \le R$ gives $\int_{|k|\le R} k^2 |\widehat{u_\varepsilon}|^2\, dk \to \int_{|k|\le R} k^2 |\widehat{u}|^2\, dk$. Letting $R \to \infty$ with $\int k^2 |\widehat{u}|^2\, dk = +\infty$ gives the conclusion. □

With these three inputs in place, the obstruction is now a direct computation.

Theorem 4.4 (L^2-divergence of the first-chaos projection)**.** *Let $f \in \mathscr{S}(\mathbb{R})$, $x \in \mathbb{R}$, $t > 0$. Assume the nondegeneracy condition*

$$C(\eta) \;:=\; \mathbb{E}[f'(x + X_\eta)] \;=\; (P_\eta f')(x), \qquad \eta > 0, \tag{4.2}$$

is not identically zero on $(0, \infty)$ (in particular, this holds whenever $f'(x) \neq 0$, since $C(\eta) \to f'(x)$ as $\eta \to 0^+$). Then

$$\liminf_{\varepsilon \to 0} \operatorname{Var}\big(\Pi_1^X C_{Z,\varepsilon}(f;t)\big) \;=\; +\infty, \tag{4.3}$$

and in particular $C_{Z,\varepsilon}(f;t)$ does not converge in $L^2(\Omega)$ as $\varepsilon \to 0$.

Proof. Set $U_r = X_r^\varepsilon$, $V_r = \partial_r X_r^\varepsilon$, $W_r = \partial_r^2 X_r^\varepsilon$. In the bulk ($r > c\varepsilon$), $\sigma_V^2 = c_\rho^{(1)}/\varepsilon$, $\operatorname{Cov}(U_r, V_r) = \frac{1}{2}$, $\operatorname{Cov}(U_r, W_r) = -c_\rho^{(1)}/\varepsilon$ (via (2.7)). Malliavin kernels: $D_w V_r = \rho_\varepsilon(r - w)$, $D_w W_r = \rho_\varepsilon'(r - w)$, $D_w U_r = \xi_{r,\varepsilon}(w)$.

Step 1: first-chaos kernel decomposition. By theorem 4.2, the conditional Malliavin kernel of $\Pi_1^X C_{Z,\varepsilon}$ is $K_\varepsilon(w \mid Y) = \mathbb{E}_X[D_w^X C_{Z,\varepsilon} \mid Y]$, splitting along the two summands of $C_{Z,\varepsilon}$:

$$K_\varepsilon(w \mid Y) \;=\; K_\varepsilon^{\mathrm{int}}(w \mid Y) \;+\; K_\varepsilon^{\mathrm{bdy}}(w \mid Y), \tag{4.4}$$

$K_\varepsilon^{\mathrm{bdy}}(w \mid Y) = L_t^0(Y)\, \mathbb{E}_X[D_w g_\varepsilon'(0)]$, $K_\varepsilon^{\mathrm{int}}(w \mid Y) = \frac{1}{2}\mathbb{E}_X[\int_0^t D_w g_\varepsilon''(|Y_s|)\, ds \mid Y]$.

Step 2: spatial support of $K_\varepsilon^{\mathrm{bdy}}$. From $g_\varepsilon'(0) = f'(x + U_0)V_0$, $\mathbb{E}_X[D_w g_\varepsilon'(0)] = \mathbb{E}_X[f''(x + U_0)V_0]D_w U_0 + \mathbb{E}_X[f'(x + U_0)]D_w V_0$. Both kernels are supported in $\{|w| \le c\varepsilon\}$: $D_w V_0 = \rho_\varepsilon(w)$ has compact support there; and $D_w U_0 = \xi_{0,\varepsilon}(w) = \int \rho_\varepsilon(z) \operatorname{sgn}(z) \mathbf{1}_{w \in [0,z]_*}\, dz$ vanishes for $|w| > c\varepsilon$ because $\rho_\varepsilon(z)$ is supported in $|z| \le c\varepsilon$. Hence

$$\operatorname{supp} K_\varepsilon^{\mathrm{bdy}}(\cdot \mid Y) \;\subseteq\; [-c\varepsilon, c\varepsilon], \qquad \forall\, \varepsilon > 0. \tag{4.5}$$

Step 3: interior kernel via Stein/Malliavin. For $g_\varepsilon''(r) = f''(x + U_r)V_r^2 + f'(x + U_r)W_r$,

$$\begin{aligned} \mathbb{E}[D_w g_\varepsilon''(r)] = \big(&\mathbb{E}[f'''(x + U_r)V_r^2] + \mathbb{E}[f''(x + U_r)W_r]\big)\xi_{r,\varepsilon}(w) \\ &+ 2\mathbb{E}[f''(x + U_r)V_r]\, \rho_\varepsilon(r - w) + \mathbb{E}[f'(x + U_r)]\, \rho_\varepsilon'(r - w). \end{aligned} \tag{4.6}$$

Stein for (U, V) jointly Gaussian: $\mathbb{E}[h(x + U)V] = \operatorname{Cov}(U, V)\mathbb{E}[h'(x + U)]$ and $\mathbb{E}[h(x + U)V^2] = \sigma_V^2 \mathbb{E}[h(x+U)] + \operatorname{Cov}(U, V)^2 \mathbb{E}[h''(x+U)]$. In the bulk, $\mathbb{E}[f'''(x+U_r)V_r^2] + \mathbb{E}[f''(x+U_r)W_r] = (\sigma_V^2 + \operatorname{Cov}(U_r, W_r))\mathbb{E}[f'''(x + U_r)] + \frac{1}{4}\mathbb{E}[f^{(5)}(x + U_r)] = \frac{1}{4}\mathbb{E}[f^{(5)}(x + U_r)]$, and $2\mathbb{E}[f''(x + U_r)V_r] = \mathbb{E}[f'''(x + U_r)]$, so the two ε^{-1} contributions cancel. Writing $A_\varepsilon(r) = \frac{1}{4}\mathbb{E}[f^{(5)}(x + U_r)]$, $B_\varepsilon(r) = \mathbb{E}[f'''(x + U_r)]$, $C_\varepsilon(r) = \mathbb{E}[f'(x + U_r)]$, with limits A, B, C given by $A(r) = \frac{1}{4}(P_{|r|} f^{(5)})(x)$, $B(r) = (P_{|r|} f''')(x)$, $C(r) = (P_{|r|} f')(x)$ matching (4.2),

$$\mathbb{E}[D_w g_\varepsilon''(r)] = A_\varepsilon(r)\xi_{r,\varepsilon}(w) + B_\varepsilon(r)\rho_\varepsilon(r - w) + C_\varepsilon(r)\rho_\varepsilon'(r - w). \tag{4.7}$$

The boundary-layer corrections at $r \in [0, c\varepsilon]$ play no role in the localised argument below. Setting $L := L_t(|Y|)$ (supported in $[0, M_t]$), integration against $L\, dr$ gives

$$K_\varepsilon^{\mathrm{int}}(w \mid Y) = \tfrac{1}{2}\big(T_1 + T_2 + T_3\big)(w), \tag{4.8}$$

where $T_1(w) = \int_0^\infty A_\varepsilon(r)L(r)\xi_{r,\varepsilon}(w)\, dr$, $T_2(w) = (\rho_\varepsilon * B_\varepsilon L)(w)$, $T_3(w) = -\partial_w(\rho_\varepsilon * C_\varepsilon L)(w)$.

Step 4: spatial localisation away from the boundary. By (4.2) and analyticity of $C(\eta)$ on $(0, \infty)$, the zero set of C is discrete; choose $J = [a, b] \subset (0, \infty)$ with $\inf_J |C| \ge c_J > 0$, and a non-empty closed subinterval $J^\circ = [a', b'] \subset (a, b)$. Fix smooth cutoff $\chi \in C_c^\infty(\mathbb{R})$ with $\chi \equiv 1$ on J°, $\operatorname{supp}\chi \subset J$. Restrict to ε with $c\varepsilon < a/2$. By Step 2, $\operatorname{supp} K_\varepsilon^{\mathrm{bdy}} \cap \operatorname{supp}\chi = \emptyset$, so $\chi K_\varepsilon^{\mathrm{bdy}} \equiv 0$ and

$$\operatorname{Var}_X(\Pi_1^X C_{Z,\varepsilon} \mid Y) = \|K_\varepsilon(\cdot \mid Y)\|_{L^2(\mathbb{R})}^2 \;\ge\; \tfrac{1}{4}\|\chi(T_1 + T_2 + T_3)\|_{L^2}^2. \tag{4.9}$$

Step 5: $\|\chi T_1\|$ and $\|\chi T_2\|$ are bounded uniformly. $|\chi| \le 1$ gives $\|\chi T_j\| \le \|T_j\|$. $\|T_1\|^2 \le (\|A\|_\infty t)^2(M_t + c\varepsilon)$ via the covariance bound and $\int L = t$; $\|T_2\|^2 \le \|B\|_\infty^2 \|L\|_\infty^2 M_t$ by Young's inequality. Both bounds are uniform in ε.

Step 6: $\|\chi T_3\| \to \infty$ on $\{M_t > b\}$. Set $u_\varepsilon = \rho_\varepsilon * (C_\varepsilon L)$. Heat-semigroup time-continuity gives $C_\varepsilon \to C$ uniformly on compacts of $(0,\infty)$; L has compact support in $[0, M_t]$; so $u_\varepsilon \to CL$ in $L^2(\mathbb{R})$, hence $\chi u_\varepsilon \to \chi CL$ in $L^2(\mathbb{R})$. Each χu_ε is in $C^1 \cap H^1$.

Claim: $\chi CL \notin H^1(\mathbb{R})$ a.s. on $\{M_t > b\}$. On J°, $\chi \equiv 1$ and $|C| \ge c_J > 0$; if $\chi CL \in H^1(\mathbb{R})$, then $CL|_{J^\circ} \in H^1_{\mathrm{loc}}(J^\circ)$, and dividing by smooth nonvanishing C gives $L \in H^1_{\mathrm{loc}}(J^\circ)$. Theorem 4.1(ii) on $J^\circ \subset (0,\infty)$ yields the contradiction $L \notin H^1_{\mathrm{loc}}(J^\circ)$ a.s. on $\{M_t > b'\} \supseteq \{M_t > b\}$.

Theorem 4.3 applied to $u_\varepsilon^\chi = \chi u_\varepsilon \to \chi CL \notin H^1(\mathbb{R})$ on $\{M_t > b\}$ gives $\liminf_{\varepsilon\to 0} \|(\chi u_\varepsilon)'\|_{L^2} = +\infty$ a.s. on $\{M_t > b\}$. Now $\chi T_3 = -(\chi u_\varepsilon)' + \chi' u_\varepsilon$ with $\|\chi' u_\varepsilon\|_{L^2} \to \|\chi' CL\|_{L^2} < \infty$; hence $\|\chi T_3\| \to +\infty$ a.s. on $\{M_t > b\}$.

Step 7: conclusion via Fatou. By the reverse triangle inequality on (4.9) and Steps 5–6, $\|K_\varepsilon(\cdot \mid Y)\|_{L^2(\mathbb{R})} \ge \frac{1}{2}(\|\chi T_3\| - \|\chi T_1\| - \|\chi T_2\|) \to +\infty$ a.s. on $\{M_t > b\}$, an event of positive $\mathbb{P}_Y$-probability (since $\{Y_t > b\} \subseteq \{M_t > b\}$). Theorem 4.2 and Fatou under $\mathbb{E}_Y$ give $\liminf_\varepsilon \mathbb{E}_Y \operatorname{Var}_X(\Pi_1^X C_{Z,\varepsilon} \mid Y) = +\infty$; the law of total variance yields (4.3). □

Remark 4.5 (Anatomy of the obstruction)**.** The proof exhibits two structural facts. (a) The naive ε^{-1}-divergent first-chaos contributions from $f''(x+U_r)V_r^2$ and $f'(x+U_r)W_r$ cancel in the bulk Malliavin kernel (4.7): this is the Stein cancellation. (b) The residual is the T_3 summand, a ρ'_ε-kernel with coefficient $C_\varepsilon L$; its $L^2(\mathbb{R})$ blow-up is forced by $\chi CL \notin H^1(\mathbb{R})$, in turn forced by the Bouleau–Yor spatial quadratic variation of the reflected Brownian local time. The first-chaos divergence is therefore not an artifact of mollification but a structural feature of the composite scenery-at-clock object.

Remark 4.6 (Wick/Hermite is not a stabilising renormalisation)**.** The diagonal contraction $\mathrm{Contr}_\varepsilon(r) = \mathbb{E}[V_r^2] = c_\rho^{(1)}/\varepsilon$ is the obvious renormalisation candidate, equivalent to Wick/Hermite square ordering in the Gaussian source atlas. The subtraction removes the even-sector diagonal contraction in $\mathcal{H}_0^X \oplus \mathcal{H}_2^X$, but it neither touches the T_3 first-chaos singularity (carried by $W_r = \partial_r^2 X^\varepsilon \in \mathcal{H}_1^X$ with $\mathrm{Var} \sim \varepsilon^{-3}$) nor preserves the Stein cancellation that isolates it: the counterterm introduced by the subtraction is itself a large first-chaos $\xi_{r,\varepsilon}$-contribution. Scalar Wick-square subtraction therefore does not remove the first-chaos obstruction and cannot by itself provide the stabilising renormalisation of the full correction. The detailed analysis is in section A.

Theorem 4.4 rules out the raw $L^2(\Omega)$ limit of the unmollified pathwise correction. A natural target is the antidual of a smooth-test subspace in a rigged Hilbert hierarchy on the second-order Malliavin energy of the IBM source; constructing the full dual concretely is open (theorem 7.1). The first-scenery-chaos slice of that dual is built in section 5: the negative-Sobolev block $H^{-1}(\mathbb{R})$ on the scenery variable hosts the limit kernel of $\Pi_1^X C_{Z,\varepsilon}$.

5 First-chaos dual closure

The obstruction of section 4 localises the divergent content of $\Pi_1^X C_{Z,\varepsilon}$ on the scenery variable $w \in \mathbb{R}$ and identifies the offending kernel as $-\partial_w(\rho_\varepsilon * C_\varepsilon L)$, which has no $L^2(\mathbb{R})$ limit. Dropping one spatial derivative absorbs the blow-up: the limit lives in $H^{-1}(\mathbb{R})$. We establish this in this section. The qualitative H^{-1} convergence (theorems 5.9, 5.12 and 5.13) is unconditional in $f \in \mathscr{S}(\mathbb{R})$ for the first-chaos sector; cross-sector $\varepsilon \to 0$ summability is open (theorem 7.1). A quantitative $\varepsilon^{1/2-\eta}$ rate (theorem 5.11) requires a Marcus–Rosen-type moment estimate (theorem 5.8) on the reflected local time, stated as a hypothesis of the proposition.

The section opens with the minimal operator-derivative bridge that names the dual object, then proves the closure in three parts: the kernel decomposition and its boundary-layer residual; the interior H^{-1} convergence (qualitative and conditional rate); the boundary contribution and the combined closure.

The operator-derivative bridge.

This subsection introduces the minimum operator-derivative structure needed to name the dual object hosting the $\varepsilon \to 0$ limit of $\Pi_1^X C_{Z,\varepsilon}$: an integration-by-parts identity fixing notation, the negative-Sobolev block on the scenery factor that will host the limit kernel, and the density question that the construction reframes. The abstract operator-derivative framework (adjointness, cylindrical reduction, chain and Leibniz rules) is developed in [4] and cited contextually; the convergence theorems below are stated and proved using joint Malliavin calculus directly.

On the product Wiener space $(\Omega, \mathcal{F}, \mathbb{P})$ with source σ-algebra $\mathcal{F}_{\mathsf{S}} = \mathcal{F}_X \vee \mathcal{F}_Y$ and joint Skorohod divergence $\delta_{\mathsf{S}}(u_X, u_Y) = \delta_X u_X + \delta_Y u_Y$ on $\mathrm{Dom}(\delta_X) \oplus \mathrm{Dom}(\delta_Y)$, the joint Malliavin derivative $D_{\mathsf{S}} = (D_X, D_Y)$ is the Hilbert adjoint of δ_{S} in the energy space $\mathcal{E}_{\mathsf{S}} = L^2(\Omega; L^2(\mathbb{R}) \oplus L^2(\mathbb{R}_+))$, the defining identity being the integration by parts

$$\mathbb{E}[F \cdot \delta_{\mathsf{S}} u] = \mathbb{E}[\langle D_{\mathsf{S}} F, u \rangle_{\mathcal{E}_{\mathsf{S}}}] \tag{5.1}$$

for $F \in \mathbb{D}^{1,2}$ on each Gaussian factor and $u \in \mathrm{Dom}(\delta_{\mathsf{S}})$. In the joint-Gaussian setting at hand D_X, D_Y are the classical Malliavin derivatives [7, 5] and (5.1) is classical; we record it only to fix notation.

The obstruction (4.3) is localised on the scenery variable $w \in \mathbb{R}$ (theorem 4.5); the natural test/dual block is therefore the standard Sobolev pair on that factor.

Definition 5.1 (First-chaos test space, dual, and generalized first-chaos functional)**.** *Set*

$$\mathcal{E}^+_{Z,1} \;\coloneqq\; L^2(\Omega_Y; H^1(\mathbb{R}, dw)), \qquad \mathcal{E}^-_{Z,1} \;\coloneqq\; L^2(\Omega_Y; H^{-1}(\mathbb{R}, dw)),$$

with the fibrewise duality pairing $\langle K(\cdot \mid Y), \Phi(\cdot \mid Y) \rangle_{H^{-1}, H^1}$ extending the $L^2(\mathbb{R})$ inner product on each fibre. A generalized Y-conditioned first-chaos functional *is an object specified by a kernel $K \in \mathcal{E}^-_{Z,1} = L^2(\Omega_Y; H^{-1}(\mathbb{R}))$; it acts on Y-random test integrands $\Phi \in \mathcal{E}^+_{Z,1} = L^2(\Omega_Y; H^1(\mathbb{R}))$ by the* first-chaos pairing

$$\big\langle K, \Phi \big\rangle_{1,*} \;\coloneqq\; \mathbb{E}_Y\big[\langle K(\cdot \mid Y), \Phi(\cdot \mid Y) \rangle_{H^{-1}, H^1}\big], \tag{5.2}$$

finite by Cauchy–Schwarz in $L^2(\Omega_Y)$. A deterministic test $\phi \in H^1(\mathbb{R})$ is the special case $\Phi(Y, w) = \phi(w)$. When $K(\cdot \mid Y) \in L^2(\mathbb{R})$ a.s., the pairing reduces to the fibrewise $L^2(\mathbb{R})$-pairing and the functional is realised by the ordinary first-chaos random variable $I_1^X(K(\cdot \mid Y)) \in L^2(\Omega)$; when $K(\cdot \mid Y)$ is only in $H^{-1}(\mathbb{R})$, no such $L^2(\Omega_X)$ random variable exists, and the functional is specified by (5.2) *alone, on Y-random test kernels Φ identified fibrewise with first-chaos test variables $I_1^X(\Phi(\cdot \mid Y))$.*

Remark 5.2 (Why H^{-1} is the natural choice)**.** Two minimal facts are available. (a) $\partial_w \colon L^2(\mathbb{R}) \to H^{-1}(\mathbb{R})$ is bounded with norm ≤ 1, since for $\phi \in H^1(\mathbb{R})$, $|\langle \partial_w f, \phi \rangle| = |\langle f, \phi' \rangle| \le \|f\|_{L^2} \|\phi\|_{H^1}$. (b) The offending kernel $T_3^\varepsilon = -\partial_w(\rho_\varepsilon * C_\varepsilon L)$ from the proof of theorem 4.4 satisfies $\rho_\varepsilon * C_\varepsilon L \to CL$ in $L^2(\mathbb{R})$ a.s., so $\partial_w(\rho_\varepsilon * C_\varepsilon L) \to \partial_w(CL)$ in $H^{-1}(\mathbb{R})$ a.s. Together these close the singular channel in H^{-1}. The sharper local threshold is governed by Bouleau–Yor regularity: $L \in H^{1/2-\eta}_{\mathrm{loc}}$ a.s., so $\partial_w(CL) \in H^{-1/2-\eta}_{\mathrm{loc}} \subset H^{-1}_{\mathrm{loc}}$. We work in H^{-1} as the natural Sobolev-dual block; no claim of optimality is made.

Remark 5.3 ("Quartic" vs. Wiener chaos degree)**.** A label sometimes attached to $\mathcal{E}^-_Z$ in this literature is "quartic dual," referring to the order of operator composition that produced the kernel: the second derivative $D^{(2)}_{\mathsf{S}}$ acts twice in assembling g''_ε from f and

the scenery, and the test side requires four-fold structure. This is not the Wiener-chaos degree of the resulting random variable. The singular object of theorem 4.4 is genuinely first-X-chaos; the first-chaos sector is a single contraction channel of the four-fold composite, not a four-variable kernel space. We use this convention throughout.

The density question. Theorem 5.1 is on the scenery side. On the observable side, a natural *candidate* intrinsic IBM derivative is obtained by joint-Malliavin compression on the maximal pullback core $\mathcal{C}_Z = \{F \in L^2(\mathcal{F}_Z) : \mathfrak{i}F \in \mathrm{Dom}(D_X) \cap \mathrm{Dom}(D_Y)\}$: write $D_Z F \coloneqq (D_X \mathfrak{i}F, D_Y \mathfrak{i}F)$ for $F \in \mathcal{C}_Z$, which is closed on this core. When the core is dense in $L^2(\mathcal{F}_Z)$, the candidate operator admits the usual Hilbert adjoint $\delta_Z = D_Z^*$, expected to satisfy the projected divergence formula $\delta_Z u = \mathbb{E}[\delta_X u_X + \delta_Y u_Y \mid \mathcal{F}_Z]$ on the lifted source-divergence domain. In the present IBM setting we do not assert such density; accordingly, the candidate intrinsic adjoint/divergence should be understood as part of the open rigged-Hilbert-space problem, not as an object used below. The abstract framework is developed in [4].

We would like to point out the following.

Open problem 5.4 (F3: density of $\mathcal{C}_Z$)**.** We do not assert density of $\mathcal{C}_Z$ in $L^2(\mathcal{F}_Z)$. The construction suggests density may fail, because the natural observables of IBM involve pointwise evaluation of the scenery at the clock $X_{|Y_t|}$, which lies outside the Sobolev domain of D_X in any unmollified sense; the mollified surrogates $X^\varepsilon_{|Y_t|}$ lie on the joint Sobolev domain at every $\varepsilon > 0$, but they are not evidently representable as $\mathcal{F}_Z$-measurable observables, and in any case the resulting approximating class is not shown here to be closed under the pullback to $\mathcal{C}_Z$. Under this scenario the natural home for the unmollified pathwise correction $C_Z(f;t) = \lim_{\varepsilon\to 0} C_{Z,\varepsilon}(f;t)$ is not $L^2(\mathcal{F}_Z)$.

The reframe is therefore the following. The candidate intrinsic IBM derivative D_Z is closed on its maximal pullback core $\mathcal{C}_Z$; we are not suggesting $\mathcal{C}_Z$ is dense in $L^2(\mathcal{F}_Z)$, and we expect it is not, for the structural reason above. The unmollified pathwise correction $C_Z(f;t)$ is consequently not expected to live in $L^2(\Omega)$: theorem 4.4 rules out its first-chaos component on classical L^2 grounds, and the natural home is a dual / rigged-Hilbert-space object. The first-chaos slice of that dual is what section 5 constructs; the full cross-sector construction is open (theorem 7.1). The label "quartic observable energy" used in this literature refers to the operator-composition order of the test side, not to chaos degree, and is not invoked structurally in any proof below.

Theorem 5.13 of section 5 proves the corresponding first-chaos limit in the dual topology that the obstruction selects, which gives the reframe constructive content. Cross-sector summability is the remaining frontier (theorem 7.1). Throughout the rest of the paper, "operator derivative" is invoked terminologically only, as the bridge that names the H^{-1} dual block; no chain rule, cylindrical reduction, or second-order $D_Z^{(2)}$ is asserted or needed in sections 5 and 6.

The kernel decomposition.

We re-derive the first-chaos Malliavin kernel of $\Pi_1^X C_{Z,\varepsilon}(f;t)$ from first principles. Recall $U_r = X^\varepsilon_r$, $V_r = \partial_r U_r$, $W_r = \partial_r^2 U_r$ with white-noise representations (2.4). We use $c_\rho^{(1)} = \int \rho^2$ and $c_\rho^{(2)} = \int (\rho')^2$ as in section 2, together with $C_\rho = \int |v|\rho(v)\,dv$ and $K_\rho = \frac{1}{2}\iint \rho(v_1)\rho(v_2)|v_1 - v_2|\,dv_1 dv_2$.

The Gaussian Stein identities (re-derived): $\mathbb{E}[h(U)V] = \mathrm{Cov}(U,V)\mathbb{E}[h'(U)]$; $\mathbb{E}[h(U)V^2] = \mathrm{Cov}(U,V)^2\mathbb{E}[h''(U)] + \sigma_V^2\mathbb{E}[h(U)]$ for jointly centred Gaussian (U,V) and smooth h.

Applying D_w to $g_\varepsilon''(r) = f''(x+U_r)V_r^2 + f'(x+U_r)W_r$ and taking $\mathbb{E}_X$ gives, in the bulk $r > c\varepsilon$ where (2.7) holds, the cancellation already used in theorem 4.4: writing $A_\varepsilon(r) = \frac{1}{4}\mathbb{E}_X[f^{(5)}(x+U_r)]$, $B_\varepsilon(r) = \mathbb{E}_X[f'''(x+U_r)]$, $C_\varepsilon(r) = \mathbb{E}_X[f'(x+U_r)]$ (bounded

uniformly in (r,ε) with heat-semigroup limits $A(r)=\frac{1}{4}(P_{|r|}f^{(5)})(x)$, $B(r)=(P_{|r|}f''')(x)$, $C(r)=(P_{|r|}f')(x)$ as in (4.2)),

$$\mathbb{E}_X[D_w g_\varepsilon''(r)] = A_\varepsilon(r)\xi_{r,\varepsilon}(w) + B_\varepsilon(r)\rho_\varepsilon(r-w) + C_\varepsilon(r)\rho_\varepsilon'(r-w). \tag{5.3}$$

Lemma 5.5 (Boundary covariance vanishes)**.** *For two-sided Brownian scenery X and even mollifier ρ,* $\mathrm{Cov}(U_0,V_0)=\frac{1}{2}(\sigma_\varepsilon^2)'(0)=0$, *hence* $U_0\perp V_0$.

Proof. $(\sigma_\varepsilon^2)'(r)=I_1'(r)-\frac{1}{2}I_2'(r)$ with $I_1(r)=\int\rho_\varepsilon(r-z)|z|\,dz$ and $I_2(r)=\iint\rho_\varepsilon(r-z_1)\rho_\varepsilon(r-z_2)|z_1-z_2|\,dz_1dz_2$. At $r=0$, translation gives $I_1'(0)=\int\rho_\varepsilon'(z)|z|\,dz=0$ by odd–even pairing. By symmetry in (z_1,z_2), $I_2'(0)=2\iint\rho_\varepsilon'(z_1)\rho_\varepsilon(z_2)|z_1-z_2|\,dz_1dz_2$; integration by parts in z_1 (compact support of ρ_ε kills the boundary) gives $I_2'(0)=-2\iint\rho_\varepsilon(z_1)\rho_\varepsilon(z_2)\,\mathrm{sgn}(z_1-z_2)\,dz_1dz_2=0$ by antisymmetry of $\mathrm{sgn}(z_1-z_2)$ against the symmetric product measure. Joint Gaussianity and vanishing covariance give independence. □

Remark 5.6 (Boundary kernel coefficient)**.** The consequence for the boundary kernel is immediate. Because $V_0\perp U_0$, $\mathbb{E}_X[f''(x+U_0^\varepsilon)V_0]=\mathbb{E}[V_0]\,\mathbb{E}_X[f''(x+U_0^\varepsilon)]=0$; the $\xi_{0,\varepsilon}$ piece in $\mathbb{E}_X[D_w g_\varepsilon'(0)]$ therefore vanishes identically, leaving only the ρ_ε-kernel with coefficient $C_\varepsilon(0)$. This rules out the naively conjectured δ_0' term in the limit of $K_\varepsilon^{\mathrm{bdy}}$ (see theorem 5.12).

Combining the Stein audit with the occupation-density formula $\int_0^t h(|Y_s|)\,ds=\int_0^\infty h(r)L_t^r(|Y|)\,dr$ ($L\coloneqq L_t(|Y|)$), the conditional Malliavin kernel of $\Pi_1^X C_{Z,\varepsilon}(f;t)$ is

$$K_\varepsilon(w\mid Y)=K_\varepsilon^{\mathrm{bdy}}(w\mid Y)+\tfrac{1}{2}\big(T_1^\varepsilon+T_2^\varepsilon+T_3^\varepsilon\big)(w\mid Y)+R_\varepsilon^{\mathrm{layer}}(w\mid Y), \tag{5.4}$$

with

$$K_\varepsilon^{\mathrm{bdy}}(w\mid Y)=L_t^0(Y)\cdot C_\varepsilon(0)\,\rho_\varepsilon(w), \tag{5.5}$$

$$T_1^\varepsilon(w\mid Y)=\int_0^\infty A_\varepsilon(r)L(r)\,\xi_{r,\varepsilon}(w)\,dr, \tag{5.6}$$

$$T_2^\varepsilon(w\mid Y)=\big(\rho_\varepsilon * B_\varepsilon L\big)(w), \tag{5.7}$$

$$T_3^\varepsilon(w\mid Y)=-\partial_w\big(\rho_\varepsilon * C_\varepsilon L\big)(w), \tag{5.8}$$

where T_j^ε use the bulk-extrapolated coefficients integrated over all $[0,\infty)$, and the boundary-layer residual

$$R_\varepsilon^{\mathrm{layer}}(w\mid Y)=\tfrac{1}{2}\int_0^{c\varepsilon}\big[\Delta_A(r)\,\xi_{r,\varepsilon}(w)+\Delta_B(r)\,\rho_\varepsilon(r-w)\big]L(r)\,dr \tag{5.9}$$

captures the deviation from the bulk identity on $r\in[0,c\varepsilon]$, with $\Delta_A(r)=(\mathrm{Var}(V_r)+\mathrm{Cov}(U_r,W_r))\mathbb{E}_X[f'''(x+U_r)]+(\mathrm{Cov}(U_r,V_r)^2-\frac{1}{4})\mathbb{E}_X[f^{(5)}(x+U_r)]$, $\Delta_B(r)=(2\,\mathrm{Cov}(U_r,V_r)-1)\mathbb{E}_X[f'''(x+U_r)]$. We write $\mathrm{Var}(V_r)$ here rather than σ_V^2 to emphasise that the boundary-layer factor must be read as the pointwise variance; the two coincide in the bulk by (2.7). Both Δ_A, Δ_B are supported in $[0,c\varepsilon]$ by (2.7); the boundary covariance $\mathrm{Cov}(U_0,V_0)=0$ (theorem 5.5) is used in (5.5) to kill the $\xi_{0,\varepsilon}$ contribution at $r=0$.

The boundary-layer residual $R_\varepsilon^{\mathrm{layer}}$ captures the deviation from the bulk Stein identity on the $O(\varepsilon)$-neighbourhood of the origin. It vanishes in the L^2 topology with an explicit $\varepsilon^{1/2}$ rate; this is the input the interior convergence theorem needs.

Lemma 5.7 (Boundary-layer residual vanishing)**.** *For every $f\in\mathscr{S}(\mathbb{R})$,*

$$R_\varepsilon^{\mathrm{layer}}\ \xrightarrow[\varepsilon\to0]{}\ 0\quad \textit{in } L^2\big(\Omega_Y;\,L^2(\mathbb{R},dw)\big),$$

and hence in $L^2(\Omega_Y;H^{-1}(\mathbb{R}))$ via the continuous embedding $L^2\hookrightarrow H^{-1}$. Moreover the bound

$$\|R_\varepsilon^{\mathrm{layer}}\|_{L^2(\mathbb{R})}\ \le\ C(f,\rho)\,\varepsilon^{1/2}\,\|L\|_\infty\quad \textit{a.s. in } Y \tag{5.10}$$

holds with $C(f,\rho)$ depending only on $\|f'''\|_\infty,\|f^{(5)}\|_\infty$, c, $\|\rho\|_\infty$, and $\int\rho^2$.

Proof. Step 1: closed form for $(\sigma_\varepsilon^2)''$. Change of variables in $\sigma_\varepsilon^2(r) = \iint \rho_\varepsilon(r-z_1)\rho_\varepsilon(r-z_2)\operatorname{Cov}(X_{z_1}, X_{z_2})\,dz_1dz_2$ (using two-sided scenery covariance) yields $\sigma_\varepsilon^2(r) = (\rho_\varepsilon * |\cdot|)(r) - K_\rho\varepsilon$, so $(\sigma_\varepsilon^2)'(r) = (\rho_\varepsilon * \operatorname{sgn})(r)$ and $(\sigma_\varepsilon^2)''(r) = 2\rho_\varepsilon(r)$, using $(\operatorname{sgn})' = 2\delta_0$ distributionally.

Step 2: layer factor bounds. Differentiating $\operatorname{Var}(U_r) = \sigma_\varepsilon^2(r)$ twice gives $2(\operatorname{Var}(V_r) + \operatorname{Cov}(U_r, W_r)) = (\sigma_\varepsilon^2)''(r)$, so $|\operatorname{Var}(V_r) + \operatorname{Cov}(U_r, W_r)| = \rho_\varepsilon(r) \le \|\rho\|_\infty/\varepsilon$ for $r \in [0, c\varepsilon]$, vanishing in the bulk. Similarly $2\operatorname{Cov}(U_r, V_r) = (\rho_\varepsilon * \operatorname{sgn})(r) \in [-1, 1]$, equal to 0 at $r = 0$ (theorem 5.5); for $r \in [0, c\varepsilon]$, $\operatorname{Cov}(U_r, V_r) \in [0, 1/2]$, so $|\operatorname{Cov}(U_r, V_r)^2 - \frac{1}{4}| \le \frac{1}{4}$ and $|2\operatorname{Cov}(U_r, V_r) - 1| \le 1$. With $f^{(k)}$ bounded for $f \in \mathscr{S}(\mathbb{R})$, $|\Delta_A(r)| \le C_f\rho_\varepsilon(r) + \frac{1}{4}\|f^{(5)}\|_\infty$, $|\Delta_B(r)| \le \|f'''\|_\infty$, where $C_f = \|f'''\|_\infty$.

Step 3: $L^2(\mathbb{R})$ bound. By theorem 2.2(a), $\|\xi_{r,\varepsilon}\|_{L^2} \le (c + C_\rho)^{1/2}\varepsilon^{1/2}$ for $r \in [0, c\varepsilon]$. Also $\|\rho_\varepsilon\|_{L^2} = (\int \rho^2)^{1/2}\varepsilon^{-1/2}$. Triangle inequality in $L^2(\mathbb{R}, dw)$ applied to (5.9):

$$\begin{aligned}\|R_\varepsilon^{\mathrm{layer}}\|_{L^2(\mathbb{R})} \le{}& \tfrac{1}{2}(c + C_\rho)^{1/2}\varepsilon^{1/2}\|L\|_\infty \int_0^{c\varepsilon} |\Delta_A(r)|\,dr \\ &+ \tfrac{1}{2}(\textstyle\int \rho^2)^{1/2}\varepsilon^{-1/2}\|L\|_\infty \int_0^{c\varepsilon} |\Delta_B(r)|\,dr.\end{aligned}$$

With $\int_0^{c\varepsilon} \rho_\varepsilon \le 1$ and $\int_0^{c\varepsilon} 1 = c\varepsilon$, $\int_0^{c\varepsilon} |\Delta_A| \le C_f + \frac{1}{4}\|f^{(5)}\|_\infty c\varepsilon$, $\int_0^{c\varepsilon} |\Delta_B| \le \|f'''\|_\infty c\varepsilon$. Substituting yields (5.10).

Step 4: assembly and conclusion. Squaring (5.10) and taking $\mathbb{E}_Y$, $\mathbb{E}_Y\|R_\varepsilon^{\mathrm{layer}}\|_{L^2}^2 \le C(f, \rho)^2\varepsilon\,\mathbb{E}_Y[\|L\|_\infty^2]$. $\mathbb{E}_Y[\|L\|_\infty^2] < \infty$ via $L_t^r(|Y|) = L_t^r(Y) + L_t^{-r}(Y)$ and the finite-moment estimate $\mathbb{E}[\sup_x L_t^x(Y)^p] < \infty$ ([6, Thm. 9.4.1]). Hence convergence to 0 at rate $O(\varepsilon)$, and the $L^2 \hookrightarrow H^{-1}$ embedding gives the second statement. □

Interior convergence in H^{-1}.

The qualitative H^{-1} convergence of the interior kernel is unconditional in $f \in \mathscr{S}(\mathbb{R})$. A quantitative rate, given at the end of the subsection, requires an additional Marcus–Rosen-type moment estimate on the reflected local time, isolated in the following assumption.

For a function $h\colon \mathbb{R}_+ \to \mathbb{R}$ supported in $[0, M_t]$, we interpret $\|h\|_{H^{1/2-\eta}([0,M_t])}$ as the $H^{1/2-\eta}(\mathbb{R})$-norm of the zero-extension of h to the whole real line. This is the norm in which the Plancherel-based mollifier estimate $\|\rho_\varepsilon * g - g\|_{L^2} \le C_\rho\varepsilon^s\|g\|_{H^s}$ operates downstream in theorem 5.11.

Assumption 5.8 (Local-time Sobolev moments)**.** *For every $\eta \in (0, 1/2)$ and $p \ge 2$, $\mathbb{E}_Y\big[\|L_t(|Y|)\|_{H^{1/2-\eta}([0,M_t])}^p\big] < \infty$, with the random-interval norm read as the whole-line $H^{1/2-\eta}(\mathbb{R})$-norm of the zero-extension as above.*

Theorem 5.8 is consistent with standard Marcus–Rosen local-time moment estimates (cf. [6]); we isolate it as an assumption because the precise Sobolev-moment formulation, read against the whole-line zero-extension above, is not needed for qualitative convergence. The qualitative convergence theorems 5.9, 5.12 and 5.13 does not invoke theorem 5.8; only the rate theorem 5.11 does.

Define the *bulk-extrapolated interior kernel* and the *actual interior kernel*:

$$K_\varepsilon^{\mathrm{int}} \coloneqq \tfrac{1}{2}(T_1^\varepsilon + T_2^\varepsilon + T_3^\varepsilon), \qquad K_\varepsilon^{\mathrm{int,exact}} \coloneqq K_\varepsilon^{\mathrm{int}} + R_\varepsilon^{\mathrm{layer}}.$$

Theorem 5.9 (Interior first-chaos H^{-1} convergence)**.** *Let $f \in \mathscr{S}(\mathbb{R})$. Then both $K_\varepsilon^{\mathrm{int}}$ and $K_\varepsilon^{\mathrm{int,exact}}$ converge to the same limit*

$$K_\varepsilon^{\mathrm{int}},\, K_\varepsilon^{\mathrm{int,exact}} \xrightarrow[\varepsilon\to 0]{} K^{\mathrm{int}} \quad \text{in } L^2\big(\Omega_Y;\, H^{-1}(\mathbb{R}, dw)\big),$$

with

$$K^{\mathrm{int}}(w \mid Y) = \tfrac{1}{2}\big[T_1^*(w \mid Y) + B(w)L(w) - \partial_w\big(C(w)L(w)\big)\big], \tag{5.11}$$

where $T_1^*(w \mid Y) = \mathbf{1}_{w\geq 0}\int_w^{M_t} A(r)L(r)\,dr$, $L = L_t(|Y|)$, $M_t = \sup_{[0,t]}|Y_s|$, *and* A, B, C *are the heat-semigroup limits of* $A_\varepsilon, B_\varepsilon, C_\varepsilon$.

Theorem 5.9 does not use the nondegeneracy hypothesis (4.2); convergence holds for every $f \in \mathscr{S}(\mathbb{R})$. The role of (4.2) is to certify that the present H^{-1}-convergent sector is the same sector whose $L^2(\Omega)$-variance diverges in theorem 4.4: under (4.2), $C \not\equiv 0$ on $(0,\infty)$, so the $T_3^* = -\partial_w(CL)$ summand is non-trivial, and the divergence in (4.3) is the $L^2(\mathbb{R})$-norm failure of the same summand. Without (4.2) the limit (5.11) still exists in H^{-1}; it is simply not witnessed by an L^2 obstruction.

Proof. By the triangle inequality, $K_\varepsilon^{\mathrm{int,exact}} - K^{\mathrm{int}} = (K_\varepsilon^{\mathrm{int}} - K^{\mathrm{int}}) + R_\varepsilon^{\mathrm{layer}}$, and $R_\varepsilon^{\mathrm{layer}} \to 0$ in $L^2(\Omega_Y; L^2(\mathbb{R})) \hookrightarrow L^2(\Omega_Y; H^{-1}(\mathbb{R}))$ by theorem 5.7. It suffices to prove $K_\varepsilon^{\mathrm{int}} \to K^{\mathrm{int}}$ in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$, using $\|\cdot\|_{H^{-1}} \leq \|\cdot\|_{L^2}$.

Decompose $K_\varepsilon^{\mathrm{int}} - K^{\mathrm{int}} = \frac{1}{2}\sum_{j=1}^3 \Delta_j^\varepsilon$ with $\Delta_j^\varepsilon = T_j^\varepsilon - T_j^*$.

(a) $T_1^\varepsilon \to T_1^*$ *in* $L^2(\Omega_Y; L^2(\mathbb{R}))$. Split $\Delta_1^\varepsilon(w) = I_\varepsilon(w) + J_\varepsilon(w)$ with $I_\varepsilon = \int_0^\infty [A_\varepsilon(r) - A(r)]L(r)\xi_{r,\varepsilon}(w)\,dr$, $J_\varepsilon = \int_0^\infty A(r)L(r)[\xi_{r,\varepsilon}(w) - \sigma_r(w)]\,dr$. L has compact support in $[0, M_t]$. Heat-semigroup time-Lipschitz (theorem 5.10 below) gives $A_\varepsilon \to A$ uniformly on $[0, M_t]$ a.s. in Y; theorem 2.2(a) gives $\|\xi_{r,\varepsilon}\|_{L^2} \leq (M_t + C_\rho)^{1/2}$. Minkowski in w: $\|I_\varepsilon\|_{L^2} \leq \|A_\varepsilon - A\|_{L^\infty([0,M_t])} \cdot t \cdot (M_t + C_\rho)^{1/2} \to 0$ a.s. in Y. For J_ε, theorem 2.2(b) gives $\|\xi_{r,\varepsilon} - \sigma_r\|_{L^2} \leq C'_\rho \varepsilon^{1/2}$, so $\|J_\varepsilon\|_{L^2} \leq \|A\|_\infty \cdot t \cdot C'_\rho \varepsilon^{1/2} \to 0$ a.s. in Y. Both bounds are dominated by $C(f,\rho)t(1 + M_t^{1/2}) \in L^2(\Omega_Y)$ (classical $\mathbb{E}_Y[M_t^p] < \infty$); dominated convergence concludes.

(b) $T_2^\varepsilon \to BL$ *in* $L^2(\Omega_Y; L^2(\mathbb{R}))$. $\Delta_2^\varepsilon = \rho_\varepsilon * ((B_\varepsilon - B)L) + (\rho_\varepsilon * BL - BL)$. Young: $\|\rho_\varepsilon * ((B_\varepsilon - B)L)\|_{L^2} \leq \|B_\varepsilon - B\|_\infty \|L\|_{L^2} \to 0$ a.s. in Y. Standard L^2-mollifier convergence: $\|\rho_\varepsilon * BL - BL\|_{L^2} \to 0$ a.s. Both terms dominated by $\|B\|_\infty \|L\|_{L^2} \in L^2(\Omega_Y)$; dominated convergence concludes.

(c) $T_3^\varepsilon \to -\partial_w(CL)$ *in* $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$. $\partial_w \colon L^2 \to H^{-1}$ has norm ≤ 1, so $\|\Delta_3^\varepsilon\|_{H^{-1}} \leq \|\rho_\varepsilon * C_\varepsilon L - CL\|_{L^2} \to 0$ a.s. in Y by the same argument as (b) applied to $g = CL$; domination by $\|C\|_\infty \|L\|_{L^2}$ finishes. □

Lemma 5.10 (Heat-semigroup time-Lipschitz)**.** *For* $h \in \mathscr{S}(\mathbb{R})$ *and* $\eta_1, \eta_2 \geq 0$, $\|P_{\eta_1} h - P_{\eta_2} h\|_{L^\infty(\mathbb{R})} \leq \frac{1}{2}|\eta_1 - \eta_2|\,\|h''\|_{L^\infty}$. *In particular* $|C_\varepsilon(r) - C(r)| \leq C_f |\sigma_\varepsilon^2(r) - |r|| \leq C_f \varepsilon$ *uniformly in* $|r| \geq c\varepsilon$, *and* $|C_\varepsilon(0) - f'(x)| \leq C_f \varepsilon$.

Refining the three Δ_j^ε estimates of theorem 5.9 with theorem 5.8 promotes the qualitative limit to a quantitative rate.

Proposition 5.11 (Quantitative interior rate, conditional)**.** *Assume theorem 5.8. Then for every* $\eta \in (0, 1/2)$,

$$\left\| K_\varepsilon^{\mathrm{int,exact}} - K^{\mathrm{int}} \right\|_{L^2(\Omega_Y; H^{-1}(\mathbb{R}))} \leq C(f,t,\rho,\eta)\,\varepsilon^{1/2-\eta}. \tag{5.12}$$

The same bound holds with $K_\varepsilon^{\mathrm{int}}$ *in place of* $K_\varepsilon^{\mathrm{int,exact}}$. *The rate is conditional on theorem 5.8; the qualitative convergence (theorem 5.9) is not.*

Proof. The residual is $O(\varepsilon^{1/2})$ in $L^2(\Omega_Y; L^2(\mathbb{R}))$ by (5.10) and $\mathbb{E}_Y[\|L\|_\infty^2] < \infty$; hence in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$ at the same rate. We refine the three Δ_j^ε estimates of theorem 5.9.

(a) Rate for Δ_1^ε. Theorem 5.10 gives $\|A_\varepsilon - A\|_{L^\infty([0,M_t])} \leq C_f \varepsilon$; theorem 2.2(b) gives $\|\xi_{r,\varepsilon} - \sigma_r\|_{L^2} \leq C'_\rho \varepsilon^{1/2}$. Combining, $\|\Delta_1^\varepsilon\|_{L^2}^2 \leq C(f,\rho)^2\, t^2(1 + M_t)\varepsilon$ a.s., so $\mathbb{E}_Y\|\Delta_1^\varepsilon\|_{L^2}^2 \leq C(f,\rho)^2 t^2(1 + \mathbb{E}_Y M_t)\varepsilon$.

(b) Rate for Δ_2^ε. $\Delta_2^\varepsilon = \rho_\varepsilon * ((B_\varepsilon - B)L) + (\rho_\varepsilon * BL - BL)$. By Young, $\|\rho_\varepsilon * ((B_\varepsilon - B)L)\|_{L^2} \leq C_f \varepsilon \|L\|_{L^2}$. For the second summand, the mollifier H^s-rate $\|\rho_\varepsilon * g - g\|_{L^2} \leq C_\rho \varepsilon^s \|g\|_{H^s}$ for $s \in (0, 1]$ (Plancherel) applied with $g = BL$, $s = 1/2 - \eta$, plus $\|BL\|_{H^{1/2-\eta}} \leq C_f \|L\|_{H^{1/2-\eta}}$ (B smooth bounded) and theorem 5.8, gives $\mathbb{E}_Y\|\Delta_2^\varepsilon\|_{L^2}^2 \leq C(f,\rho)^2 \varepsilon^{1-2\eta}$.

(c) Rate for Δ_3^ε in H^{-1}. Same argument as (b) for $g = CL$, $s = 1/2 - \eta$, then $\|\partial_w\|_{L^2 \to H^{-1}} \le 1$: $\mathbb{E}_Y \|\Delta_3^\varepsilon\|_{H^{-1}}^2 \le C(f,\rho)^2 \varepsilon^{1-2\eta}$.

$\|\cdot\|_{H^{-1}} \le \|\cdot\|_{L^2}$ on (a) and (b); summing yields (5.12). □

The rate is consistent with the sharp Brownian local-time Sobolev threshold at $1/2^-$: L has spatial Sobolev regularity exactly up to $H^{1/2-\eta}$, and the mollifier rate $\|\rho_\varepsilon * g - g\|_{L^2} \sim \varepsilon^s$ for $g \in H^s$ is sharp at the regularity index of g. The η-loss is intrinsic to Brownian local-time regularity, not an artefact of the proof. No endpoint rate is claimed. The rate depends on the moment input theorem 5.8; the qualitative convergence does not.

Boundary contribution and the combined closure.

The T_1 and T_2 pieces of K^{int} are in $L^2(\mathbb{R})$ on the qualitative limit, while the T_3 piece $-\partial_w(CL)$ is generally only in $H^{-1}(\mathbb{R})$; this is the failure identified in theorem 4.4. The boundary kernel $K_\varepsilon^{\mathrm{bdy}} = L_t^0(Y) C_\varepsilon(0) \rho_\varepsilon$ collapses to the singular distribution $L_t^0(Y) f'(x) \delta_0$. Since $\delta_0 \notin H^{-1/2}$ but $\delta_0 \in H^{-1/2-\delta}$ for every $\delta > 0$, the boundary convergence is most naturally stated one Sobolev step weaker than H^{-1}; the inclusion $H^{-1/2-\delta} \subset H^{-1}$ for $\delta \in (0, 1/2)$ then assembles the combined statement.

Theorem 5.12 (Boundary first-chaos $H^{-1/2-\delta}$ convergence)**.** *For every $f \in \mathscr{S}(\mathbb{R})$ and every $\delta \in (0, 1/2)$,*

$$K_\varepsilon^{\mathrm{bdy}} \xrightarrow[\varepsilon \to 0]{} L_t^0(Y)\, f'(x)\, \delta_0 \quad \text{in } L^2\big(\Omega_Y;\, H^{-1/2-\delta}(\mathbb{R})\big),$$

with rate $\big\|K_\varepsilon^{\mathrm{bdy}} - L_t^0(Y) f'(x) \delta_0\big\|_{L^2(\Omega_Y; H^{-1/2-\delta}(\mathbb{R}))} \le C_{f,\rho}\, \varepsilon^{\delta \wedge 1/2}$. The range $\delta \in (0, 1/2)$ matches the Fourier-estimate range used in the proof and is the range needed by theorem 5.13 for the embedding $H^{-1/2-\delta} \subset H^{-1}$. No δ_0' contribution arises (theorems 5.5 and 5.6).

Proof. $K_\varepsilon^{\mathrm{bdy}} = L_t^0(Y) C_\varepsilon(0) \rho_\varepsilon$. Split $K_\varepsilon^{\mathrm{bdy}} - L_t^0(Y) f'(x) \delta_0 = L_t^0(Y)(C_\varepsilon(0) - f'(x))\rho_\varepsilon + L_t^0(Y) f'(x)(\rho_\varepsilon - \delta_0)$.

Step 1 (coefficient rate): $C_\varepsilon(0) = (P_{\sigma_\varepsilon^2(0)} f')(x)$; $\sigma_\varepsilon^2(0) = O(\varepsilon)$ by Step 1 of theorem 5.7. Theorem 5.10 gives $|C_\varepsilon(0) - f'(x)| \le C_f \varepsilon$.

Step 2 (distribution rate): Plancherel and $\hat\rho_\varepsilon(k) = \hat\rho(\varepsilon k)$, $|\hat\rho(\varepsilon k) - 1| \le \min(2, c_\rho \varepsilon^2 k^2)$, together with the split at $|k| = 1/\varepsilon$: $\|\rho_\varepsilon - \delta_0\|_{H^{-s}} \le C\varepsilon^{s-1/2}$ for $s \in (1/2, 5/2)$. Applied at $s = 1/2 + \delta$: $\|\rho_\varepsilon - \delta_0\|_{H^{-1/2-\delta}} \le C\varepsilon^\delta$.

Step 3 (assemble): Triangle inequality and $\mathbb{E}[L_t^0(Y)^2] < \infty$ ([6, Thm. 6.1.1]) give the claimed bound, with the slower component $\varepsilon^{\delta \wedge 1/2}$. □

The Sobolev exponent in theorem 5.12 cannot be improved at the boundary: $\delta_0 \in H^s(\mathbb{R})$ if and only if $s < -1/2$, so $\delta_0 \notin H^{-1/2}$ but $\delta_0 \in H^{-1/2-\delta}$ for every $\delta > 0$. We work at $\delta \in (0, 1/2)$, which both matches the Fourier-estimate range in the proof and gives the inclusion $H^{-1/2-\delta} \subset H^{-1}$ needed to assemble the combined H^{-1} closure of the full first-X-chaos Malliavin kernel.

Corollary 5.13 (Combined first-chaos H^{-1} convergence)**.** *For every $f \in \mathscr{S}(\mathbb{R})$ and every $\delta \in (0, 1/2)$, the full first-X-chaos Malliavin kernel $K_\varepsilon = K_\varepsilon^{\mathrm{bdy}} + K_\varepsilon^{\mathrm{int,exact}}$ satisfies*

$$K_\varepsilon \xrightarrow[\varepsilon \to 0]{} K \coloneqq L_t^0(Y) f'(x) \delta_0 + K^{\mathrm{int}} \quad \text{in } L^2\big(\Omega_Y;\, H^{-1}(\mathbb{R}, dw)\big).$$

Under theorem 5.8, the combined rate is $\varepsilon^{(1/2-\eta)\wedge\delta}$ for every $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$; choosing $\delta = 1/2 - \eta$ optimises to $\varepsilon^{1/2-\eta}$.

Proof. Combine theorems 5.9 and 5.12. The embedding $H^{-1/2-\delta} \hookrightarrow H^{-1}$ for $\delta \in (0, 1/2)$ is standard; the interior $L^2 \hookrightarrow H^{-1}$ embedding is direct. □

Corollary 5.14 (First-chaos dual closure of $\Pi_1^X C_{Z,\varepsilon}$)**.** *At every $\varepsilon > 0$, $\Pi_1^X C_{Z,\varepsilon}(f;t)$ is an honest first-chaos random variable with conditional Malliavin kernel $K_\varepsilon \in L^2(\Omega_Y; L^2(\mathbb{R}))$; theorem 5.13 gives $K_\varepsilon \to K$ in $L^2(\Omega_Y; H^{-1}(\mathbb{R})) = \mathcal{E}^-_{Z,1}$, with limit kernel K generically only H^{-1} (not L^2), so the limit object is a generalized Y-conditioned first-chaos functional (theorem 5.1) rather than an $L^2(\Omega_X)$ random variable. Convergence is the duality-pairing convergence: for every $\Phi \in \mathcal{E}^+_{Z,1} = L^2(\Omega_Y; H^1(\mathbb{R}))$,*

$$\big\langle \Pi_1^X C_{Z,\varepsilon}(f;t), \Phi \big\rangle_{1,*} \xrightarrow[\varepsilon\to 0]{} \mathbb{E}_Y\big[\langle K(\cdot \mid Y), \Phi(\cdot \mid Y)\rangle_{H^{-1},H^1}\big],$$

including the special case of deterministic $\phi \in H^1(\mathbb{R})$. Under theorem 5.8, the rate is $\varepsilon^{1/2-\eta}$.

Theorems 5.13 and 5.14 close the $L^2(\Omega)$-divergent first scenery-chaos sector of $C_{Z,\varepsilon}(f;t)$ in the first-X-chaos block of the operator-derivative dual. They do not prove:

- that $C_Z(f;t)$ exists in any global cross-sector dual; the (m,n)-bidegree projections with $m \geq 2$ are not addressed here (theorem 7.1);
- that the full unmollified IBM Itô identity (1.3) survives $\varepsilon \to 0$ in any topology;
- that the Brownian-time semigroup correction $u(t,x) - f(x)$ arises from the first-chaos sector. It cannot: first chaos has mean zero, so $\mathbb{E}[\Pi_1^X C_Z(f;t)] = 0$. The semigroup correction lives in $\Pi_{0,0}$ (theorem 6.3).

Filtration-level obstructions (Tsirelson-type non-Brownian noises) are an independent direction and play no role here. The next section places this closure inside the joint Wiener bidegree lattice and shows how the same construction, at fixed ε, recovers the Allouba–Zheng semigroup correction in the pure-clock sector.

6 The bidegree atlas at fixed ε

This section is supporting structure rather than a fourth main result: it provides a fixed-ε structural map of $C_{Z,\varepsilon}(f;t)$ across the joint Wiener chaos lattice $\mathcal{H}_m^X \otimes \mathcal{H}_n^Y$, locating the first-chaos dual closure of section 5 inside the lattice and exhibiting the sector that carries the Brownian-time semigroup correction of theorem 3.6. At fixed ε, $C_{Z,\varepsilon} \in L^2(\Omega_X \times \Omega_Y)$ and the joint chaos expansion

$$C_{Z,\varepsilon}(f;t) \;=\; \sum_{m,n\geq 0} \Pi_{m,n}\, C_{Z,\varepsilon}(f;t) \tag{6.1}$$

converges in $L^2(\Omega)$ by Wiener–Itô (theorem 6.1 below). All statements that follow are at fixed $\varepsilon > 0$; cross-sector $\varepsilon \to 0$ summability is the open problem theorem 7.1. Three sectors carry the structural content: pure-clock $(0,\bullet)$, first-scenery-chaos $(1,\bullet)$, and the $(2,0)$ scalar energy-contraction channel.

Theorem 6.1 (Bidegree decomposition at every ε)**.** *For every $f \in \mathscr{S}(\mathbb{R})$, $t \geq 0$, $\varepsilon > 0$, $C_{Z,\varepsilon}(f;t) \in L^2(\Omega_X \times \Omega_Y)$ and* (6.1) *converges in $L^2(\Omega_X \times \Omega_Y)$. The decomposition applies summand-by-summand to the two terms $g'_\varepsilon(0) L^0_t(Y)$ and $\frac{1}{2}\int_0^t g''_\varepsilon(|Y_s|)\, ds$ of $C_{Z,\varepsilon}$.*

Proof. At fixed $\varepsilon > 0$, each of $g'_\varepsilon(0)$, $g''_\varepsilon(r)$ has uniform $L^2(\Omega_X)$ bounds in r (smooth bounded $f^{(k)}(x + U_r)$ for $k \leq 5$, Gaussian factors V_r, V_r^2, W_r in L^p at fixed ε); $L^0_t(Y) \in L^p(\Omega_Y)$ for every p; Minkowski preserves L^2 under $\int_0^t$. Hence $C_{Z,\varepsilon}(f;t) \in L^2(\Omega)$, and (6.1) is the orthogonal expansion in (2.3), convergent by Wiener–Itô. □

The pure-clock sector $(0,\bullet)$**.**

The pure-clock sector is X-deterministic; it carries the expectation-level content of the correction. Define the full X-expectation of the scalar integrand:

$$\psi_\varepsilon(r) \;\coloneqq\; \mathbb{E}_X[g_\varepsilon''(r)] \;=\; \mathbb{E}_X[f''(x+U_r)V_r^2 + f'(x+U_r)W_r]. \tag{6.2}$$

Stein on (U_r,V_r) and (U_r,W_r) gives

$$\psi_\varepsilon(r) \;=\; \mathrm{Cov}(U_r,V_r)^2\,\mathbb{E}_X[f^{(4)}(x+U_r)] + \big[\mathrm{Var}(V_r)+\mathrm{Cov}(U_r,W_r)\big]\,\mathbb{E}_X[f''(x+U_r)], \tag{6.3}$$

where we write $\mathrm{Var}(V_r)$ in place of σ_V^2 to keep the boundary-layer factor pointwise; the two coincide in the bulk by (2.7). By (2.7), $\mathrm{Cov}(U_r,V_r)^2=\frac14$ and $\mathrm{Var}(V_r)+\mathrm{Cov}(U_r,W_r)=0$ for $r\ge c\varepsilon$. Setting

$$\phi_\varepsilon^{\mathrm{int}}(r) \coloneqq \tfrac14\mathbb{E}_X[f^{(4)}(x+U_r)], \qquad b_\varepsilon(r)\coloneqq \psi_\varepsilon(r)-\phi_\varepsilon^{\mathrm{int}}(r), \tag{6.4}$$

the boundary residual b_ε is smooth, supported in $[0,c\varepsilon]$, and satisfies (by (6.3) and bracket bounds on the layer)

$$\sup_r |b_\varepsilon(r)| \le C_f\varepsilon^{-1}, \qquad \int_0^\infty |b_\varepsilon(r)|\,dr \le C_f, \tag{6.5}$$

with C_f depending on $\|f^{(2)}\|_\infty, \|f^{(4)}\|_\infty$ and ρ only. The bound (6.5) follows by the same bracket argument used in theorem 5.7: the boundary-layer factors $|\,\mathrm{Var}(V_r)+\mathrm{Cov}(U_r,W_r)|$ and $|\,\mathrm{Cov}(U_r,V_r)^2-\frac14|$ are bounded by $\rho_\varepsilon(r)\le\|\rho\|_\infty/\varepsilon$ and $\frac14$ respectively for $r\in[0,c\varepsilon]$, both vanishing on $\{r\ge c\varepsilon\}$.

Lemma 6.2 (Boundary Tanaka piece vanishes after $\mathbb{E}_X$)**.** $\mathbb{E}_X[g_\varepsilon'(0)]=0$.

Proof. $\mathbb{E}_X[g_\varepsilon'(0)] = \mathbb{E}_X[f'(x+U_0)V_0] = \mathrm{Cov}(U_0,V_0)\mathbb{E}_X[f''(x+U_0)] = 0$ by the scalar Stein identity and the boundary covariance $\mathrm{Cov}(U_0,V_0)=0$ (theorem 5.5). □

Theorem 6.3 (Pure-clock sector and expectation limit)**.** *For every $f\in\mathscr{S}(\mathbb{R})$, $t\ge0$, $\varepsilon>0$:*

1. *(Scalar identity)*

$$\Pi_0^X\,C_{Z,\varepsilon}(f;t) \;=\; F_\varepsilon(Y;t) \;\coloneqq\; \tfrac12\int_0^t \psi_\varepsilon(|Y_s|)\,ds,$$

   with $\psi_\varepsilon=\phi_\varepsilon^{\mathrm{int}}+b_\varepsilon$ as in (6.4)*.*
2. *(Bidegree slices) For every $n\ge0$, $\Pi_{0,n}\,C_{Z,\varepsilon}(f;t)=\Pi_n^Y F_\varepsilon(Y;t)$.*
3. *(Expectation limit, referring to theorem 3.6)*

$$\Pi_{0,0}\,C_{Z,\varepsilon}(f;t) \;=\; \mathbb{E}\big[C_{Z,\varepsilon}(f;t)\big] \;\xrightarrow{\varepsilon\to0}\; u(t,x)-f(x).$$

   The limit decomposes as

$$u(t,x)-f(x) \;=\; \underbrace{\lim_{\varepsilon\to0}\tfrac12\mathbb{E}_Y\int_0^t \phi_\varepsilon^{\mathrm{int}}(|Y_s|)\,ds}_{\text{interior bi-Laplacian shadow}} \;+\; \underbrace{\lim_{\varepsilon\to0}\tfrac12\mathbb{E}_Y\int_0^t b_\varepsilon(|Y_s|)\,ds}_{\text{boundary-layer source}};$$

   the boundary-layer piece converges in expectation to $\int_0^t \partial_x^2 f(x)/(2\sqrt{2\pi s})\,ds$. We identify this limit only at the level of expectations; see theorem 6.4.
4. *(Clock-fluctuation blocks) For $n\ge1$, $\Pi_{0,n}=\Pi_n^Y F_\varepsilon$ is centred in $\mathcal{H}_n^Y$. In particular, by Stroock–Clark–Ocone [7],*

$$\Pi_{0,1}\,C_{Z,\varepsilon}(f;t) \;=\; I_1^Y\big(\mathbb{E}_Y[D_\cdot^Y F_\varepsilon(Y;t)]\big),$$

   with $D_u^Y|Y_s| = \mathrm{sgn}(Y_s)\mathbf{1}_{u\le s}$ on the standard Bouleau–Hirsch domain.

Proof. (1) $\Pi_0^X = \mathbb{E}_X$ gives $\Pi_0^X C_{Z,\varepsilon} = \mathbb{E}_X[g_\varepsilon'(0)]L_t^0(Y) + \frac{1}{2}\int_0^t \mathbb{E}_X[g_\varepsilon''(|Y_s|) \mid Y]\,ds$. The first term is zero by theorem 6.2. Independence of X, Y gives $\mathbb{E}_X[g_\varepsilon''(|Y_s|) \mid Y] = \psi_\varepsilon(|Y_s|)$. (2) $\Pi_{0,n} = \Pi_n^Y \Pi_0^X$ applied to (1). (3) $\Pi_{0,0} = \mathbb{E}_Y \mathbb{E}_X = \mathbb{E}$. The expectation-limit step is the statement of theorem 3.6 and is not re-proved here. The interior bi-Laplacian piece is identified by $\phi_\varepsilon^{\mathrm{int}}(r) \to \frac{1}{4}(P_{|r|}f^{(4)})(x)$ uniformly on compacts of $(0,\infty)$ (heat-semigroup continuity) and dominated convergence under the half-normal law of $|Y_s|$: $\frac{1}{2}\mathbb{E}_Y\int_0^t \phi_\varepsilon^{\mathrm{int}}(|Y_s|)\,ds \to \frac{1}{8}\int_0^t \Delta^2 u(s,x)\,ds$. The boundary-layer piece limit is then identified by subtraction from theorem 3.6. (4) For $n \geq 1$, $\Pi_n^Y F_\varepsilon \in \mathcal{H}_n^Y$ is centred; Stroock–Clark–Ocone gives the $n = 1$ Malliavin-derivative representation; the absolute-value Malliavin derivative is the Bouleau–Hirsch standard form. □

Open problem 6.4 (F2: boundary-layer random functional limit)**.** Theorem 6.3(3) identifies the $\varepsilon \to 0$ limit of $\mathbb{E}_Y\int_0^t b_\varepsilon(|Y_s|)\,ds$ (an expectation-level identification of the $t^{-1/2}$ source). We make no claim about the random-functional $L^2(\Omega_Y)$ limit of $\int_0^t b_\varepsilon(|Y_s|)\,ds$ as $\varepsilon \to 0$; (6.5) shows that the mass of b_ε concentrates at scale ε near $r = 0$, suggesting a clock local-time at 0 object as the natural candidate limit, but the identification and convergence are open. Consequently the decomposition $\psi_\varepsilon = \phi_\varepsilon^{\mathrm{int}} + b_\varepsilon$ is used at the expectation level only (theorem 6.3(3)); the $(0,n)$ slices for $n \geq 1$ are recorded via the projection formula at fixed ε (theorem 6.3(2),(4)), not in their $\varepsilon \to 0$ limit.

The first-scenery-chaos sector $(1,\bullet)$.

By theorem 4.2, the first-scenery-chaos projection of $C_{Z,\varepsilon}$ satisfies $\Pi_1^X C_{Z,\varepsilon}(f;t) = I_1^X(K_\varepsilon(\cdot \mid Y))$ with K_ε given by (5.4).

Theorem 6.5 (First-scenery-chaos bidegree slices)**.** *For every $f \in \mathscr{S}(\mathbb{R})$, $t \geq 0$, $\varepsilon > 0$, $n \geq 0$:*

1. *(Block formula)* $\Pi_{1,n} C_{Z,\varepsilon}(f;t) = I_1^X(\Pi_n^Y K_\varepsilon(\cdot \mid Y))$.
2. *(Clock-averaged block, $n = 0$)* $\Pi_{1,0} C_{Z,\varepsilon}(f;t) = I_1^X(\mathbb{E}_Y K_\varepsilon(\cdot \mid Y))$. *This sector has zero full expectation (every $\mathcal{H}_1^X$ element is centred in X) but is generally non-zero as an element of $\mathcal{H}_1^X$.*
3. *(Dual closure inheritance) The dual closure of theorem 5.13 descends block-by-block: as $\varepsilon \to 0$, $\Pi_n^Y K_\varepsilon \to \Pi_n^Y K^{\lim}$ in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$, with $K^{\lim} = K^{\mathrm{int}} + L_t^0(Y)f'(x)\delta_0$. In particular $\mathbb{E}_Y K_\varepsilon \to \mathbb{E}_Y K^{\lim}$ in $H^{-1}(\mathbb{R})$. The convergence is at the level of conditional Malliavin kernels; the corresponding first-chaos objects $\Pi_{1,n} C_{Z,\varepsilon}$ are ordinary $L^2(\Omega)$-valued random variables at every fixed $\varepsilon > 0$, but their limit is the generalized Y-conditioned first-chaos functional of theorem 5.1 specified by the kernel $\Pi_n^Y K^{\lim} \in L^2(\Omega_Y; H^{-1}(\mathbb{R}))$ via the duality pairing, not an ordinary $L^2(\Omega)$-valued random variable.*

Proof. (1) Π_n^Y acts only on Ω_Y and commutes with the Ω_X-linear $I_1^X(\cdot)$: $\Pi_n^Y[I_1^X(K_\varepsilon)] = I_1^X(\Pi_n^Y K_\varepsilon)$. Π_1^X is the identity on first-X-chaos. (2) $n = 0$ specialisation: $\Pi_0^Y = \mathbb{E}_Y$, full-expectation zero by centring of $\mathcal{H}_1^X$. Non-vanishing: the FCDC limit kernel $K^{\lim}$ has explicit non-trivial Y-average (e.g. via $-\partial_w(C(w)L_t^w(|Y|))$ with $\mathbb{E}_Y[L_t^w(|Y|)]$ not identically zero). (3) Π_n^Y is an orthogonal projection on $L^2(\Omega_Y)$, hence a contraction on $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$. Apply it to $K_\varepsilon \to K^{\lim}$ in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$ (theorem 5.13). The $n = 0$ specialisation gives the $\mathbb{E}_Y$ statement. □

Remark 6.6 (Boundary local time inside Π_1^X)**.** The boundary kernel $K_\varepsilon^{\mathrm{bdy}} \to L_t^0(Y)f'(x)\delta_0$ in $H^{-1/2-\delta}(\mathbb{R}) \subset H^{-1}(\mathbb{R})$, so the first-chaos boundary singularity is a feature of Π_1^X, distributed across the $\Pi_{1,n}$ blocks via the Y-chaos expansion of $L_t^0(Y)$. The expectation $\mathbb{E}_Y[L_t^0(Y)] = \sqrt{2t/\pi}$ contributes to $\Pi_{1,0}$; the higher Y-chaos pieces of $L_t^0(Y)$ contribute to $\Pi_{1,n}$ for $n \geq 1$.

The energy-contraction channel $(2,0)$**.**

By theorem 6.1 and partial expectations, $\Pi_{2,0}\, C_{Z,\varepsilon}(f;t) = \mathbb{E}_Y[\Pi_2^X C_{Z,\varepsilon}(f;t)]$.

Gaussian Stroock and product formula. $V_r \in \mathcal{H}_1^X$ with $V_r^2 = \sigma_V^2 + I_2^X(\rho_\varepsilon(r-\cdot)^{\otimes 2})$, the second summand being the Wick-square realisation of the second-energy contraction (we use this only as Gaussian-coordinate bookkeeping). $W_r = I_1^X(\rho_\varepsilon'(r-\cdot))$ is first-chaos. For $r \geq 0$ and $k \in \mathbb{N}$, $f^{(k)}(x+U_r)$ admits the Stroock expansion $f^{(k)}(x+U_r) = \sum_{m\geq 0} I_m^X(h_{m,r,\varepsilon}^{(k)})$ with $h_{m,r,\varepsilon}^{(k)} = \frac{1}{m!}\mathbb{E}_X[f^{(k+m)}(x+U_r)]\xi_{r,\varepsilon}^{\otimes m}$. The Gaussian product formula $I_p^X(a) I_q^X(b) = \sum_{j=0}^{p\wedge q} j!\binom{p}{j}\binom{q}{j} I_{p+q-2j}^X(a \otimes_j b)$ ([7, §1.1.3]) isolates the chaos-2 selector $p+q-2j=2$.

Proposition 6.7 (Channel enumeration)**.** *For every fixed $\varepsilon > 0$ and $r \geq 0$,*

$$\Pi_2^X[f''(x+U_r)V_r^2] = \mathsf{C}_{0,0} + \mathsf{C}_{2,V} + \mathsf{C}_{2,\mathrm{c}} + \mathsf{C}_4,$$
$$\Pi_2^X[f'(x+U_r)W_r] = \mathsf{C}_1 + \mathsf{C}_3,$$

where (with $h_{m,r}^{(k)} = h_{m,r,\varepsilon}^{(k)}$):

$$\begin{aligned}
\mathsf{C}_{0,0} &\coloneqq \mathbb{E}_X[f''(x+U_r)] \cdot I_2^X(\rho_\varepsilon(r-\cdot)^{\otimes 2}),\\
\mathsf{C}_{2,V} &\coloneqq \sigma_V^2 \cdot I_2^X(h_{2,r}^{(2)}),\\
\mathsf{C}_{2,\mathrm{c}} &\coloneqq 4 \cdot I_2^X(h_{2,r}^{(2)} \otimes_1 \rho_\varepsilon(r-\cdot)^{\otimes 2}),\\
\mathsf{C}_4 &\coloneqq 12 \cdot I_2^X(h_{4,r}^{(2)} \otimes_2 \rho_\varepsilon(r-\cdot)^{\otimes 2}),\\
\mathsf{C}_1 &\coloneqq I_2^X(h_{1,r}^{(1)} \mathbin{\widetilde{\otimes}} \rho_\varepsilon'(r-\cdot)),\\
\mathsf{C}_3 &\coloneqq 3 \cdot I_2^X(h_{3,r}^{(1)} \otimes_1 \rho_\varepsilon'(r-\cdot)).
\end{aligned}$$

There are no further chaos-2 contributions.

Proof. Direct substitution of the Stroock expansion into Π_2^X and applying the product formula. The parity constraint $p+q \equiv 2 \pmod 2$ and the index bound $j \leq \min(p,q)$ enumerate all allowed (p,q,j). Coefficients $j!\binom{p}{j}\binom{q}{j}$ evaluate to the listed values. □

Theorem 6.8 ($(2,0)$ energy-contraction channel decomposition)**.** *For every $f \in \mathscr{S}(\mathbb{R})$, $t \geq 0$, $\varepsilon > 0$,*

$$\Pi_{2,0}\, C_{Z,\varepsilon}(f;t) \;=\; \sqrt{\tfrac{2t}{\pi}}\, \Pi_2^X[f'(x+U_0)V_0] + \tfrac{1}{2}\int_0^t \mathbb{E}_Y\big[\Pi_2^X g_\varepsilon''(|Y_s|)\big]\, ds, \tag{6.6}$$

with the second-chaos integrand given by theorem 6.7 as $\mathsf{C}_{0,0}+\mathsf{C}_{2,V}+\mathsf{C}_{2,\mathrm{c}}+\mathsf{C}_4+\mathsf{C}_1+\mathsf{C}_3$ at $r=|Y_s|$. The expression lies in $\mathcal{H}_2^X \otimes \mathcal{H}_0^Y$ at every $\varepsilon > 0$.

Proof. By $\Pi_{2,0} = \mathbb{E}_Y \Pi_2^X$ applied to (3.4). For the boundary term, $g_\varepsilon'(0)$ is Y-independent, so $\Pi_2^X[g_\varepsilon'(0) L_t^0(Y)] = L_t^0(Y)\Pi_2^X[f'(x+U_0)V_0]$, and $\mathbb{E}_Y[L_t^0(Y)] = \sqrt{2t/\pi}$ by Tanaka and the half-normal distribution of $|Y_t|$. The interior term is direct from theorem 6.7 at $r=|Y_s|$. □

Proposition 6.9 (Boundary $(2,0)$ kernel)**.** $\Pi_2^X[f'(x+U_0)V_0] = I_2^X\big(h_{1,0,\varepsilon}^{(1)} \mathbin{\widetilde{\otimes}} \rho_\varepsilon(0-\cdot)\big)$.

Proof. $V_0 = I_1^X(\rho_\varepsilon(0-\cdot)) \in \mathcal{H}_1^X$, so $q=1$. The parity analysis of theorem 6.7 with $q=1$ allows $p \in \{1,3\}$ with coefficients 1 and 3 respectively. The $p=1, j=0$ channel contributes the displayed first term. The $p=3, j=1$ channel is formally allowed by parity, but its 1-fold contraction $h_{3,0,\varepsilon}^{(1)} \otimes_1 \rho_\varepsilon(0-\cdot)$ contains the inner product

$$\int_{\mathbb{R}} \xi_{0,\varepsilon}(u)\, \rho_\varepsilon(0-u)\, du \;=\; \int_{\mathbb{R}} (D_u U_0)(D_u V_0)\, du \;=\; \mathrm{Cov}(U_0, V_0) \;=\; 0$$

by theorem 5.5 (the same identity that kills the δ_0' boundary contribution in theorem 5.12). Hence the $p = 3, j = 1$ channel vanishes identically, leaving only the $p = 1, j = 0$ term. □

Remark 6.10 (The $(2,0)$ block is potentially ε-singular)**.** The leading scalar channel $\mathsf{C}_{0,0}$ involves the kernel $\rho_\varepsilon(r-\cdot)^{\otimes 2}$, whose $L^2(\mathbb{R}^2)$ norm is $O(\varepsilon^{-1})$; multiplied by the $O(1)$ scalar $\mathbb{E}_X[f''(x+U_r)]$, this channel is potentially singular as $\varepsilon \to 0$. Whether an energy-contraction subtraction stabilises it is part of the cross-sector Tier-A problem (theorem 7.1). We do not provide a proof of any specific $\varepsilon \to 0$ rate for the $(2,0)$ sector; the decomposition theorem 6.8 is a fixed-ε structural statement only.

table 1 summarises the atlas at fixed ε. Three structural distinctions are worth restating:

- *Deterministic mass vs. zero expectation.* Only $\Pi_{0,0}$ has non-zero full expectation. Every other $\Pi_{m,n}$ with $m+n \geq 1$ is centred. "Zero expectation" is strictly weaker than "zero bidegree"; the prototypical example is $\Pi_{1,0} = I_1^X(\mathbb{E}_Y K_\varepsilon)$, generally non-zero (theorem 6.5).
- *Two distinct boundary objects.* The $(0,0)$ boundary residual b_ε is an X-expectation-level object producing the $t^{-1/2}$ source at the level of $\mathbb{E}$ (theorem 6.3(3)); the first-chaos boundary local time $L_t^0(Y)f'(x)\delta_0$ is a Y-randomness-level object inside Π_1^X (theorems 5.12 and 6.6). They are not the same and live in different bidegree rows.
- *The $(2,0)$ contraction channel does not remove the $(1,\bullet)$ singularity.* The six-channel decomposition is in $\mathcal{H}_2^X$; the obstruction is in $\mathcal{H}_1^X$. Wick-square renormalisation acts on $V_r^2 - \sigma_V^2$ and leaves the first-chaos rank-one W_r-singularity untouched (theorem 4.6).

Table 1: Bidegree atlas of $C_{Z,\varepsilon}(f;t)$ at fixed ε.

Sector	**Object**	**Status**
$(0,0)$, interior of ψ_ε	bi-Laplacian shadow	explicit (theorem 6.3)
$(0,0)$, boundary residual b_ε	singular $t^{-1/2}$ source	expectation limit identified; random-functional limit open (theorem 6.4)
$(0,0)$, combined	deterministic mass $u(t,x) - f(x)$	theorem 3.6
$(0,n)$, $n \geq 1$	pure-clock zero-mean fluctuations	projection formula at fixed ε
$(1,0)$	clock-averaged scenery $I_1^X(\mathbb{E}_Y K_\varepsilon)$	closed in H^{-1} (theorem 5.13)
$(1,n)$, $n \geq 1$	Y-resolved singular scenery channels	closed in H^{-1} (theorem 6.5)
bdy: $L_t^0(Y)f'(x)\delta_0$	first-chaos boundary local time	spread over $(1,n)$
$(2,0)$	six-channel energy-contraction kernel	explicit at fixed ε (theorem 6.8)
$(2,n)$, $n \geq 1$	clock-resolved second-scenery channels	open (theorem 7.1)
(m,n), $m \geq 3$	higher scenery contractions	open (theorem 7.1)

Cross-sector $\varepsilon \to 0$ summability of $\sum_{m,n} \Pi_{m,n} C_{Z,\varepsilon}$ in a global dual topology would require, beyond this paper, four ingredients: a norm on the $(2,0)$ block in which the six channels (or a suitable contraction subtraction) converge; control of the mixed $(2,n)$ sectors; a tail estimate for $\Pi_{m,n}$ with $m \geq 3$; and identification of the boundary residual as an $L^2(\Omega_Y)$ limit. None of these is needed for the fixed-ε atlas above: theorems 6.1, 6.3, 6.5 and 6.8 hold at every $\varepsilon > 0$. The four ingredients are catalogued in section 7 alongside the three open problems.

7 Open problems and scope

This section collects the three named open problems that this paper exposes, together with directions for future work.

The three open problems.

(F1) *Global cross-sector $\varepsilon \to 0$ summability* (theorem 7.1 below). The atlas of section 6 is a fixed-ε structural map. We do not claim convergence of $\sum_{m,n} \Pi_{m,n} C_{Z,\varepsilon}$ as $\varepsilon \to 0$ in any global cross-sector dual topology. The three loci where $\varepsilon \to 0$ is proved are: (a) the expectation (theorem 3.6); (b) the first-scenery-chaos sector in $L^2(\Omega_Y; H^{-1}(\mathbb{R}))$ (theorem 5.13); (c) the $\Pi_{0,0}$ block, which is (a) by theorem 6.3(3).

(F2) *The boundary-layer random-functional $L^2(\Omega_Y)$ limit* (theorem 6.4). The expectation limit of $\mathbb{E}_Y \int_0^t b_\varepsilon(|Y_s|)\, ds$ is identified as the $t^{-1/2}$ source. The random-functional limit of $\int_0^t b_\varepsilon(|Y_s|)\, ds$ in $L^2(\Omega_Y)$ is open.

(F3) *Density of an intrinsic IBM core, and the natural home for $C_Z(f;t)$* (theorem 5.4). The candidate intrinsic IBM derivative is closed on its maximal pullback core $\mathcal{C}_Z$; we are not suggesting $L^2(\mathcal{F}_Z)$-density of $\mathcal{C}_Z$, and we expect it fails for the structural reason that scenery-at-clock observables lie outside the joint Sobolev domain in any unmollified sense. The natural home for the unmollified pathwise correction is therefore not $L^2(\Omega)$ but a dual / rigged-Hilbert-space object; the first-chaos slice is constructed in section 5.

Open problem 7.1 (F1: cross-sector summability)**.** For $f \in \mathscr{S}(\mathbb{R})$ satisfying (4.2), the $\varepsilon \to 0$ behaviour of $\Pi_{m,n} C_{Z,\varepsilon}$ for (m,n) with $m \geq 2$, or $m = 0$ and $n \geq 1$, is open: we have neither a uniform-in-ε bound nor an identification of an $\varepsilon \to 0$ limit in any topology. The construction of a cross-sector dual norm in which $\sum_{m,n} \Pi_{m,n} C_{Z,\varepsilon} \to C_Z(f;t)$ as $\varepsilon \to 0$ remains open; it is the principal cross-sector analytic frontier this paper exposes.

Directions for future work.

Beyond the three open problems above, the present construction opens several directions.

1. *Unmollified clock derivative.* Identify $\lim_{\varepsilon \to 0} D_Y C_{Z,\varepsilon}(f;t)$ as a distribution-valued kernel against $L^2(\mathbb{R}_+)$ test integrands. The clock component of the natural intrinsic IBM derivative is genuinely singular at the level of local time.
2. *Cylindrical reduction for D_Z.* Establish a chain or product rule for D_Z via a cylindrical reduction with random representers (cf. [4]); the natural candidate is the joint Brownian–Malliavin representer projected onto $\mathcal{F}_Z$.
3. *Second-order intrinsic energy and PDE shadow.* Construct $D_Z^{(2)}$ as a closed operator and verify that, averaged against the law of Y, the second-order Γ-form built from $D_Z^{(2)}$ matches the bi-Laplacian/local-time PDE shadow of [1]. The fixed-ε recovery in theorem 6.3(3) gives a target on the source side; the intrinsic-IBM-side recovery remains open.
4. *Second-energy-contraction sector.* Construct the $\mathcal{H}_2^X$ tensor-product Bessel-potential dual block analogous to H^{-1} and study $\varepsilon \to 0$ convergence of the six channels of theorem 6.8 or a suitable contraction-subtraction thereof.
5. *Wick/Hermite hierarchy in higher chaoses.* The $\mathcal{H}_{m\geq 2}^X$ sectors admit contraction renormalisations (Wick/Hermite in Gaussian coordinates). Establishing chaos-by-chaos L^2-boundedness uniformly in ε, then assembling across (m,n), is a natural route towards theorem 7.1.

6. *Filtration-level questions.* The natural filtration of Z is a separate matter; Tsirelson-type questions about whether $\mathcal{F}_Z$ is Brownian-generated are independent of the present construction.

A Wick / Hermite renormalisation is not a stabilising renormalisation

The first-chaos obstruction (theorem 4.4) suggests an obvious renormalisation: subtract the diagonal contraction $\mathrm{Contr}_\varepsilon(r) = c_\rho^{(1)}/\varepsilon$ from the squared first-energy direction $(\partial_r X^\varepsilon)^2$. In Gaussian source coordinates this is the Wick / Hermite square in $\mathcal{H}_2^X$. The subtraction removes the obvious even-sector diagonal contraction, but it neither touches the T_3 first-chaos singularity (carried by $\partial_r^2 X^\varepsilon$, a centred Gaussian in $\mathcal{H}_1^X$ with $\mathrm{Var} \sim \varepsilon^{-3}$) nor preserves the cancellation mechanism that isolates that singularity: the counterterm it introduces is itself a large first-chaos $\xi_{r,\varepsilon}$-contribution. Scalar Wick-square subtraction therefore does not remove the first-chaos obstruction and cannot by itself provide the stabilising renormalisation of the full correction.

The diagonal contraction is

$$\mathrm{Contr}_\varepsilon(r) \;\coloneqq\; \mathbb{E}[(\partial_r X_r^\varepsilon)^2] \;=\; \frac{c_\rho^{(1)}}{\varepsilon} \quad \text{for } |r| > c\varepsilon, \tag{A.1}$$

so Wick / Hermite centering is the Brownian-coordinate expression of an operator-energy contraction subtraction. Define

$$:(\partial_r X_r^\varepsilon)^2: \;\coloneqq\; (\partial_r X_r^\varepsilon)^2 - c_\rho^{(1)}/\varepsilon \tag{A.2}$$

and the renormalised correction candidate $\widehat{C}_{Z,\varepsilon}(f;t) \coloneqq g_\varepsilon'(0) L_t^0(Y) + \frac{1}{2}\int_0^t : g_\varepsilon''(|Y_s|) : \, ds$, where $: g_\varepsilon''(r) := f''(x+U_r) : (\partial_r U_r)^2 : + f'(x+U_r) W_r$.

Proposition A.1 (Scalar Wick subtraction does not remove the first-chaos obstruction)**.** *Under the hypotheses of theorem 4.4, the contraction subtraction* (A.2) *does not modify the $\partial_r^2 X^\varepsilon$ content of g_ε'', so the first-scenery-chaos singularity in $\Pi_1^X C_{Z,\varepsilon}$ carried by $\partial_r^2 X^\varepsilon$ (identified in theorem 4.4) is not removed. In addition, the subtraction introduces a counterterm whose first-chaos contribution is large, so the Stein cancellation that isolates the T_3 singularity is not preserved. Scalar Wick-square subtraction therefore does not remove the first-chaos obstruction and cannot by itself provide the stabilising renormalisation of the full correction.*

Proof. The subtraction (A.2) alters only the V^2 factor in $g_\varepsilon''(r) = f''(x+U_r)V_r^2 + f'(x+U_r)W_r$. The term $f'(x+U_r)W_r$ carries the $C_\varepsilon(r)\rho_\varepsilon'(r-w)$ contribution to the first-chaos Malliavin kernel (4.7), which is the source of the residual L^2-divergence in theorem 4.4; it is untouched by (A.2). The subtraction further adds a counterterm $-\frac{c_\rho^{(1)}}{2\varepsilon}\int_0^t f''(x + X^\varepsilon_{|Y_s|})\, ds$ whose first-chaos Malliavin kernel is built from the U_r-fibre $\xi_{r,\varepsilon}(w)$ (via Stein on $f''(x+U_r)$). This ε^{-1}-scaled $\xi_{r,\varepsilon}$-kernel modifies the T_1 summand of (4.7)–(4.8) and disrupts the cancellation (4.7), while leaving $T_3 = -\partial_w(\rho_\varepsilon * C_\varepsilon L)$ intact. □

The renormalisation must operate at the level of a quartic composite observable energy, not at the level of a single Gaussian square. Two analytic routes remain open: (i) interpret $\partial_r^2 X$ in the unmollified limit as a Hida/Schwartz distribution and pair $f'(X)\partial_r^2 X$ against test integrands in the negative-Sobolev dual; (ii) Stratonovich-regularise the time integral $\int_0^t f'(x + X^\varepsilon_{|Y_s|})\partial_r^2 X^\varepsilon_{|Y_s|}\, ds$ via integration by parts in the clock variable, transferring the spatial r-derivative onto Brownian-clock local-time data. Neither route is pursued here.

B Fourier-mode PDE-shadow diagnostic

We record a PDE-shadow diagnostic identifying, at the level of Fourier-mode test functions, the second-order signature reproduced by theorem 3.1.

For $k \in \mathbb{R}$ and $t > 0$, $u_k(t,x) \coloneqq \mathbb{E}[e^{ikZ_t}]$. Conditioning on $|Y_t| = s$ gives $\mathbb{E}[e^{ikX_s}] = e^{-k^2 s/2}$, and $|Y_t|$ has half-normal density $p_t(s) = \sqrt{2/(\pi t)}\, e^{-s^2/(2t)}$ for $s \geq 0$.

Proposition B.1 (Fourier-mode bi-Laplacian shadow)**.** *Let $a = k^2/2$ and $q(t) \coloneqq \int_0^\infty e^{-as} p_t(s)\, ds$, so $u_k(t,x) = e^{ikx} q(t)$. Then*

$$q'(t) \;=\; \frac{a^2}{2}\, q(t) \;-\; \frac{a}{\sqrt{2\pi t}} \;=\; \frac{k^4}{8} q(t) - \frac{k^2}{2\sqrt{2\pi t}}, \tag{B.1}$$

and equivalently $\partial_t u_k(t,x) = \frac{1}{8}\Delta^2 u_k(t,x) + \Delta f_k(x)/(2\sqrt{2\pi t})$ for $f_k(x) = e^{ikx}$.

Proof. $\phi_t(s) = (2\pi t)^{-1/2} e^{-s^2/(2t)}$ satisfies $\partial_t \phi_t = \frac{1}{2}\partial_s^2 \phi_t$; the same holds for $p_t = \sqrt{2}\phi_t|_{s\geq 0}$ at every $s > 0$. Hence $q'(t) = \frac{1}{2}\int_0^\infty e^{-as}\partial_s^2 p_t(s)\, ds$. IBP twice in s, using $\partial_s p_t(0) = 0$ and $p_t(0) = \sqrt{2/(\pi t)}$: $\int_0^\infty e^{-as}\partial_s^2 p_t\, ds = a(-p_t(0) + a q(t)) = a^2 q(t) - a\sqrt{2/(\pi t)}$. Dividing by 2 and substituting $a = k^2/2$ gives (B.1); the PDE follows from $\Delta e^{ikx} = -k^2 e^{ikx}$ and $\Delta^2 e^{ikx} = k^4 e^{ikx}$. □

Fourier-synthesis on smooth bounded f gives, with $u(t,x) = \mathbb{E}[f(x + X_{|Y_t|})]$, $\partial_t u(t,x) = \frac{1}{8}\Delta^2 u(t,x) + \Delta f(x)/(2\sqrt{2\pi t})$, the Brownian-time PDE of [1].

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References

[1] H. Allouba and W. Zheng, *Brownian-time processes: the PDE connection and the half-derivative generator*, Ann. Probab. **29** (2001), no. 4, 1780–1795.

[2] A. N. Borodin and P. Salminen, *Handbook of Brownian Motion — Facts and Formulae*, 2nd ed., Probability and Its Applications, Birkhäuser Verlag, Basel, 2002.

[3] N. Bouleau and M. Yor, *Sur la variation quadratique des temps locaux de certaines semimartingales*, C. R. Acad. Sci. Paris Sér. I Math. **292** (1981), no. 9, 491–494.

[4] R. Fontes, *The Operator Derivative in Continuous Stochastic Calculus: A Hilbert Energy Space Framework*, SSRN preprint, 2026. Available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6381359.

[5] S. Janson, *Gaussian Hilbert Spaces*, Cambridge Tracts in Mathematics, vol. 129, Cambridge University Press, Cambridge, 1997.

[6] M. B. Marcus and J. Rosen, *Markov Processes, Gaussian Processes, and Local Times*, Cambridge Studies in Advanced Mathematics, vol. 100, Cambridge University Press, 2006.

[7] D. Nualart, *The Malliavin Calculus and Related Topics*, 2nd ed., Probability and Its Applications, Springer-Verlag, Berlin, 2006.

[8] D. Revuz and M. Yor, *Continuous Martingales and Brownian Motion*, 3rd ed., Grundlehren der Mathematischen Wissenschaften, vol. 293, Springer-Verlag, Berlin, 1999.